

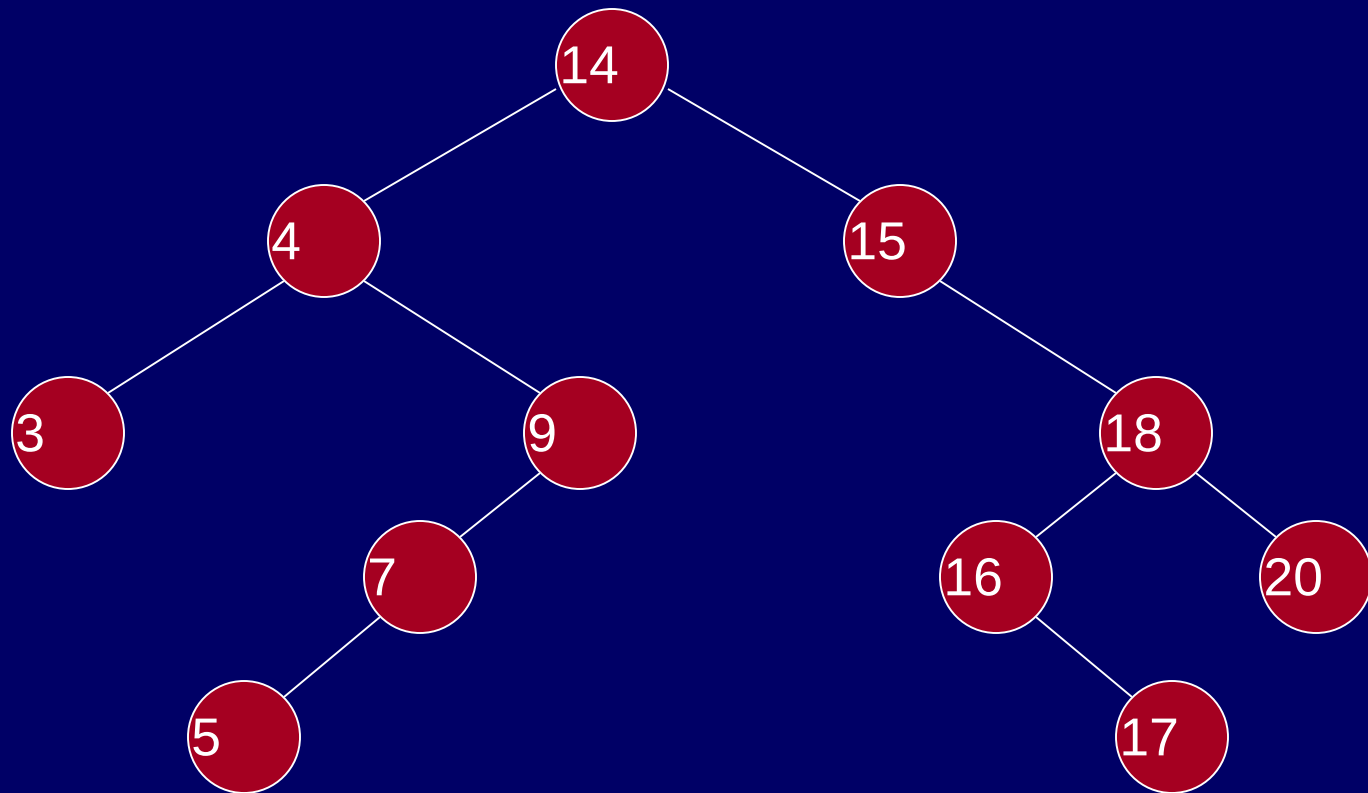
In the Name of Allah the Most
Beneficent the Most Merciful

Subject : Data Structures & Algorithms

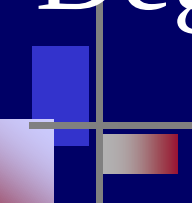
Lecture : 07

Degenerate Binary Search Tree

BST for 14, 15, 4, 9, 7, 18, 3, 5, 16, 20, 17



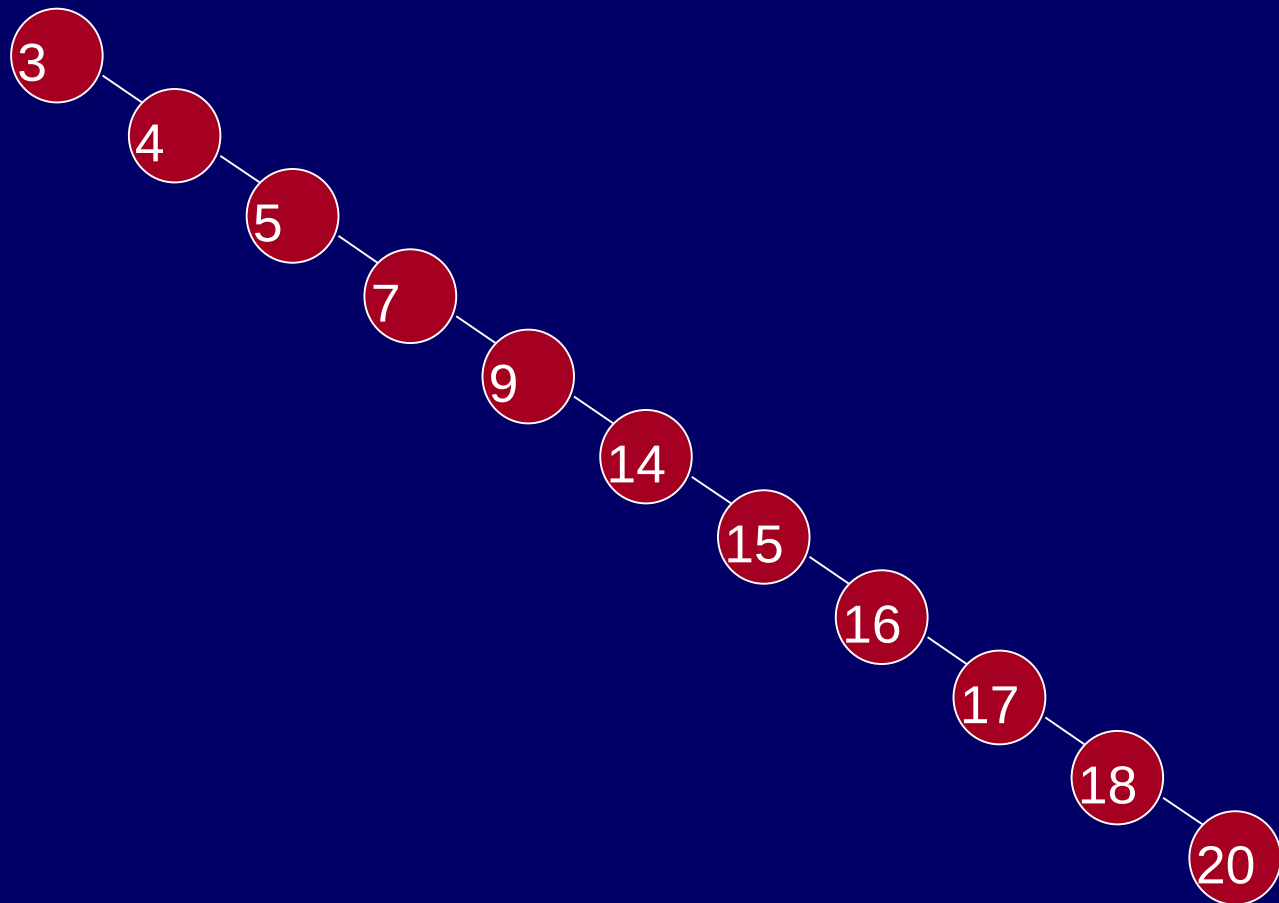
Degenerate Binary Search Tree



BST for 3 4 5 7 9 14 15 16 17 18 20

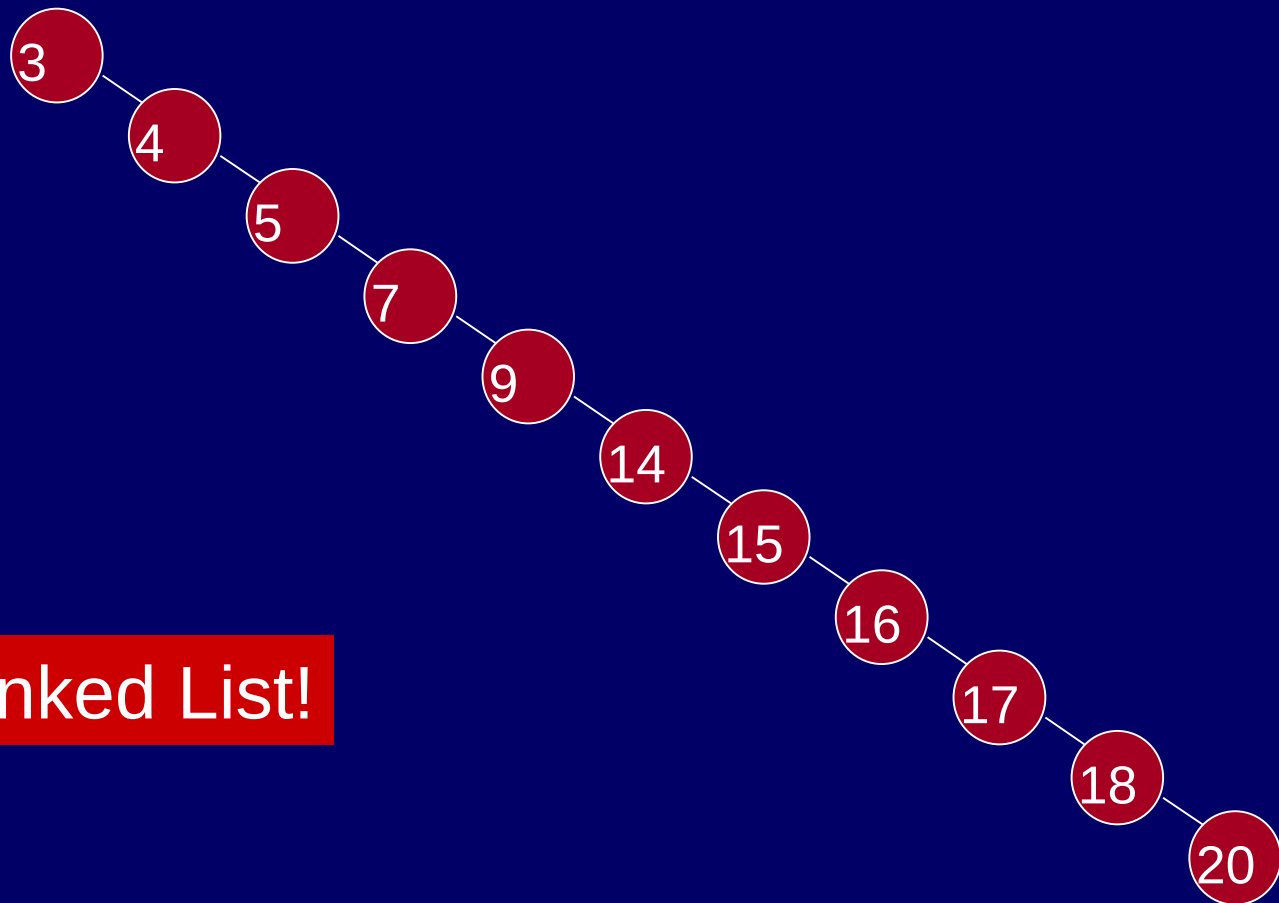
Degenerate Binary Search Tree

BST for 3 4 5 7 9 14 15 16 17 18 20



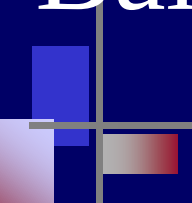
Degenerate Binary Search Tree

BST for 3 4 5 7 9 14 15 16 17 18 20



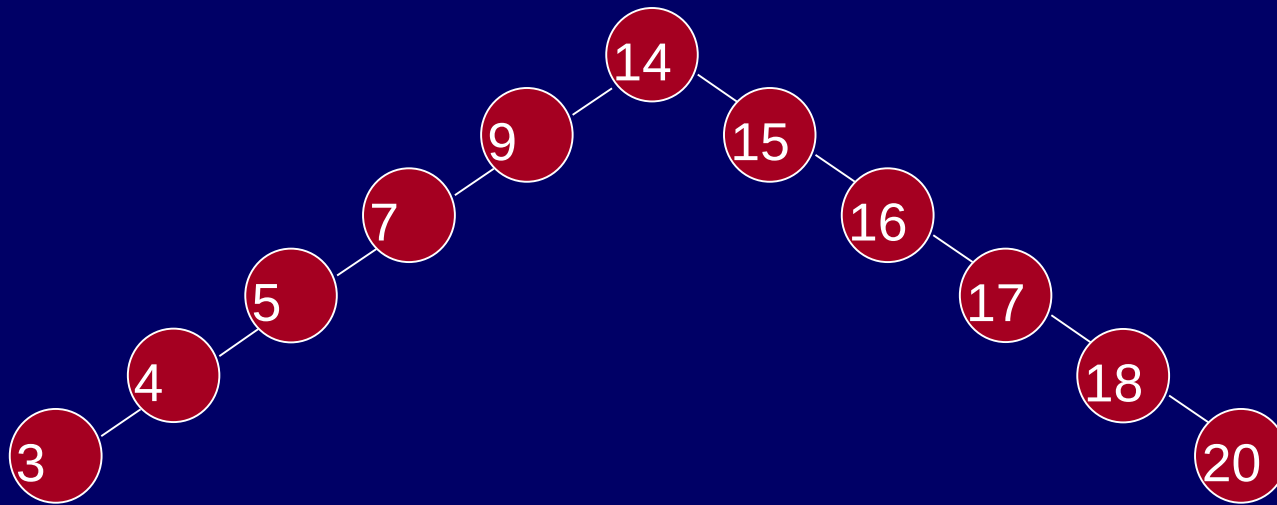
Linked List!

Balanced BST



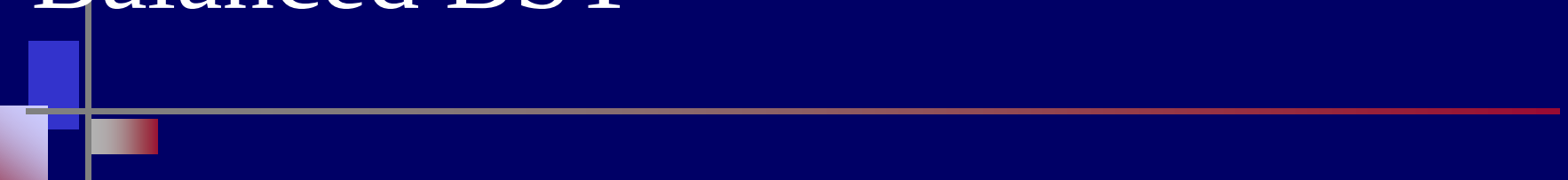
- We should keep the tree *balanced*.
- One idea would be to have the left and right subtrees have the same height

Balanced BST



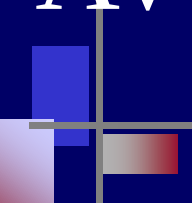
Does not force the tree to be *shallow*.

Balanced BST



- We could insist that every node must have left and right subtrees of same height.
- But this requires that the tree be a complete binary tree
- To do this, there must have $(2^{d+1} - 1)$ data items, where d is the depth of the tree.
- This is too rigid a condition.

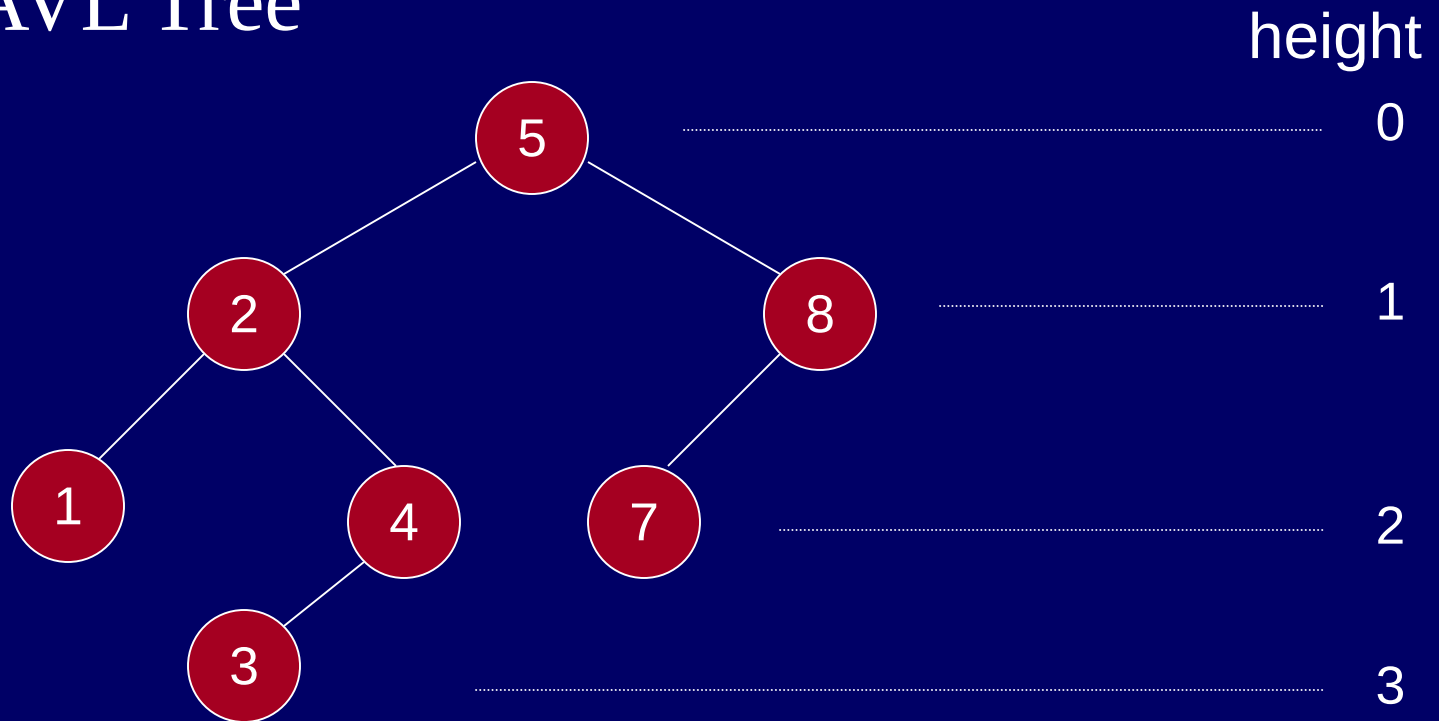
AVL Tree



- AVL (Adelson-Velskii and Landis) tree.
- An AVL tree is identical to a BST except
 - height of the left and right subtrees can differ by at most 1.
 - height of an empty tree is defined to be (-1) .

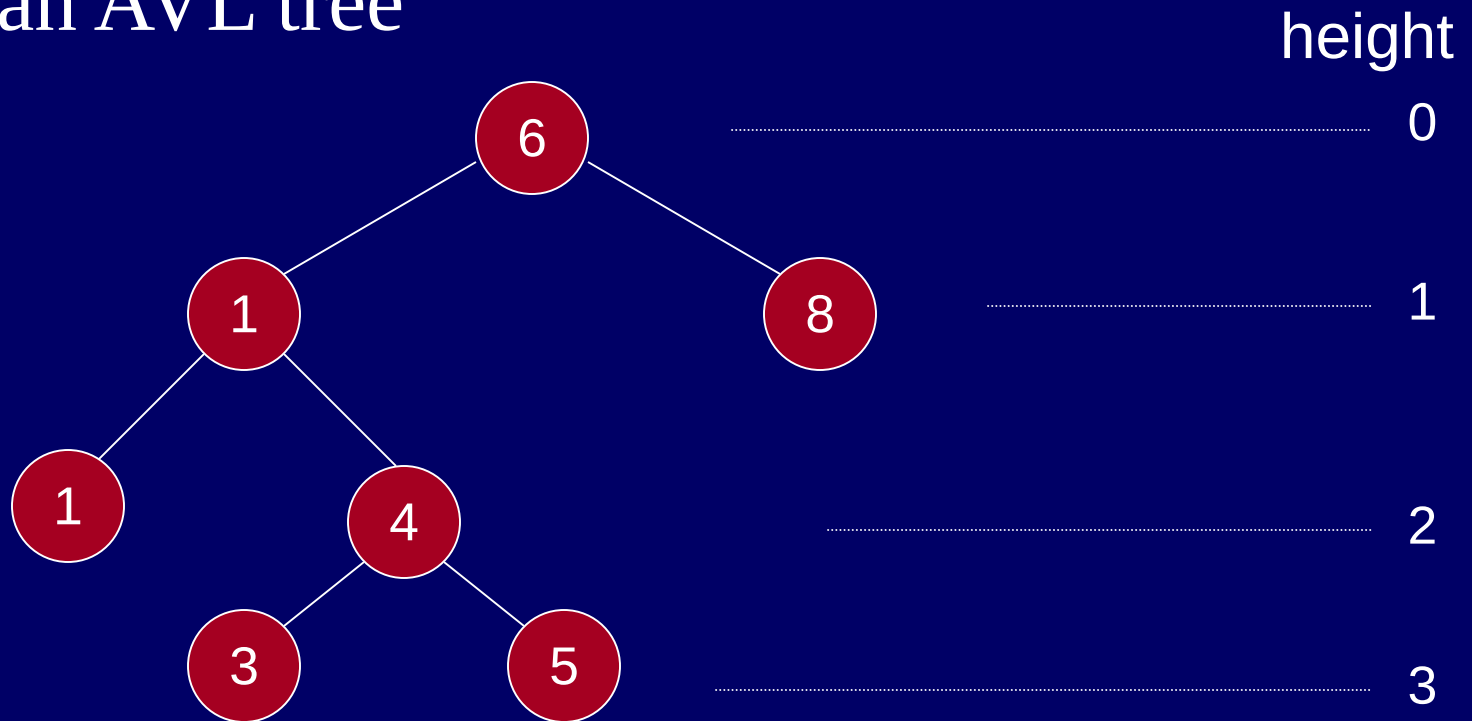
AVL Tree

- An AVL Tree

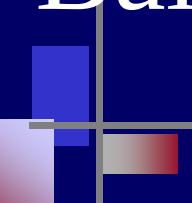


AVL Tree

- Not an AVL tree

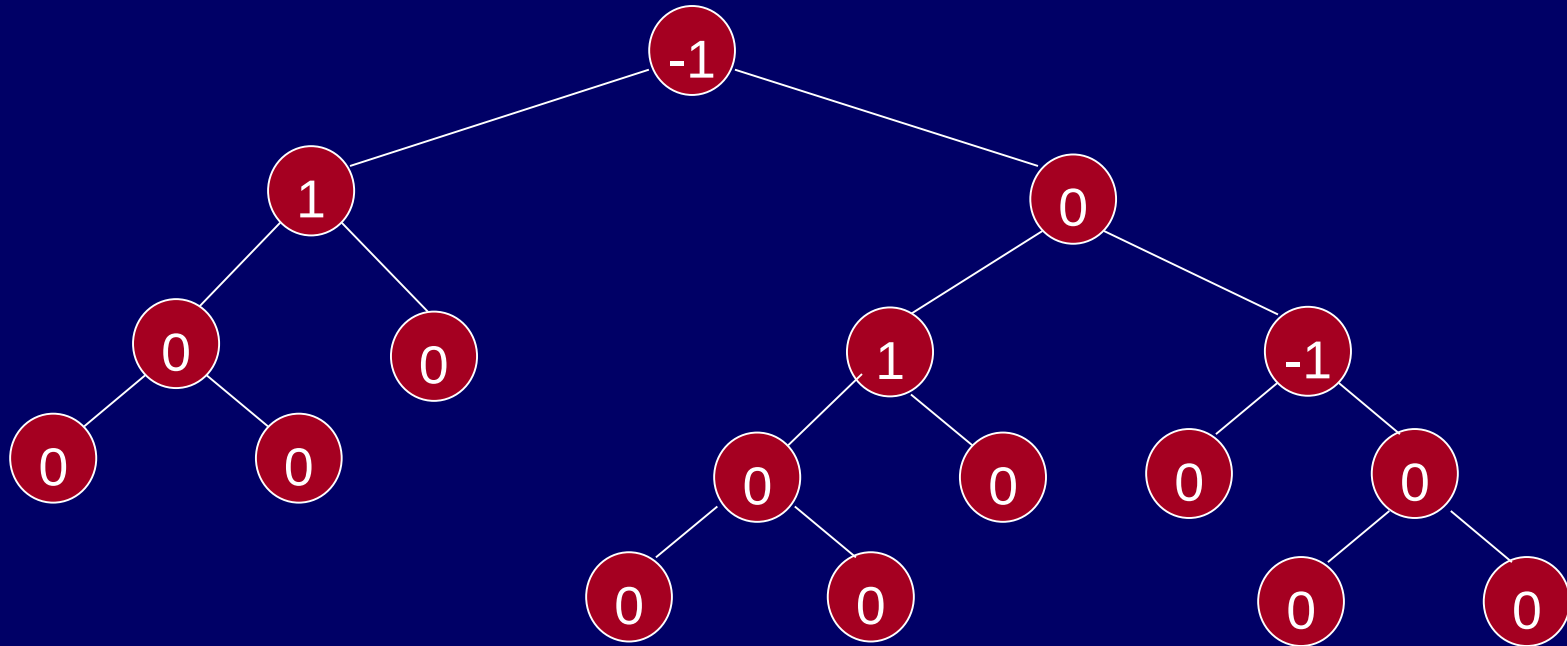


Balanced Binary Tree



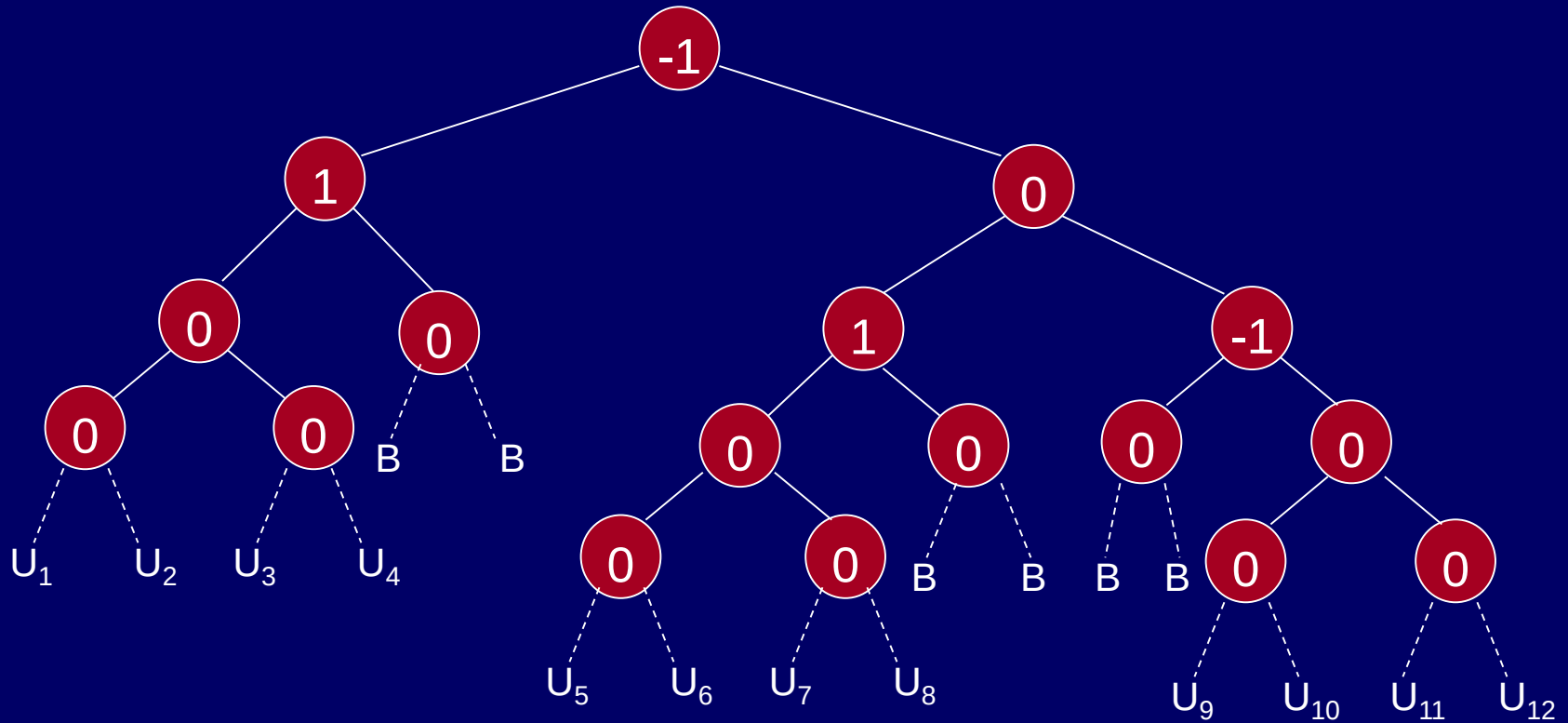
- The *height* of a binary tree is the maximum level of its leaves (also called the depth).
- The balance of a node in a binary tree is defined as the **height of its left subtree** minus **height of its right subtree**.
- Here, for example, is a balanced tree. Each node has an indicated balance of 1, 0, or -1 .

Balanced Binary Tree



Balanced Binary Tree

Insertions and effect on balance



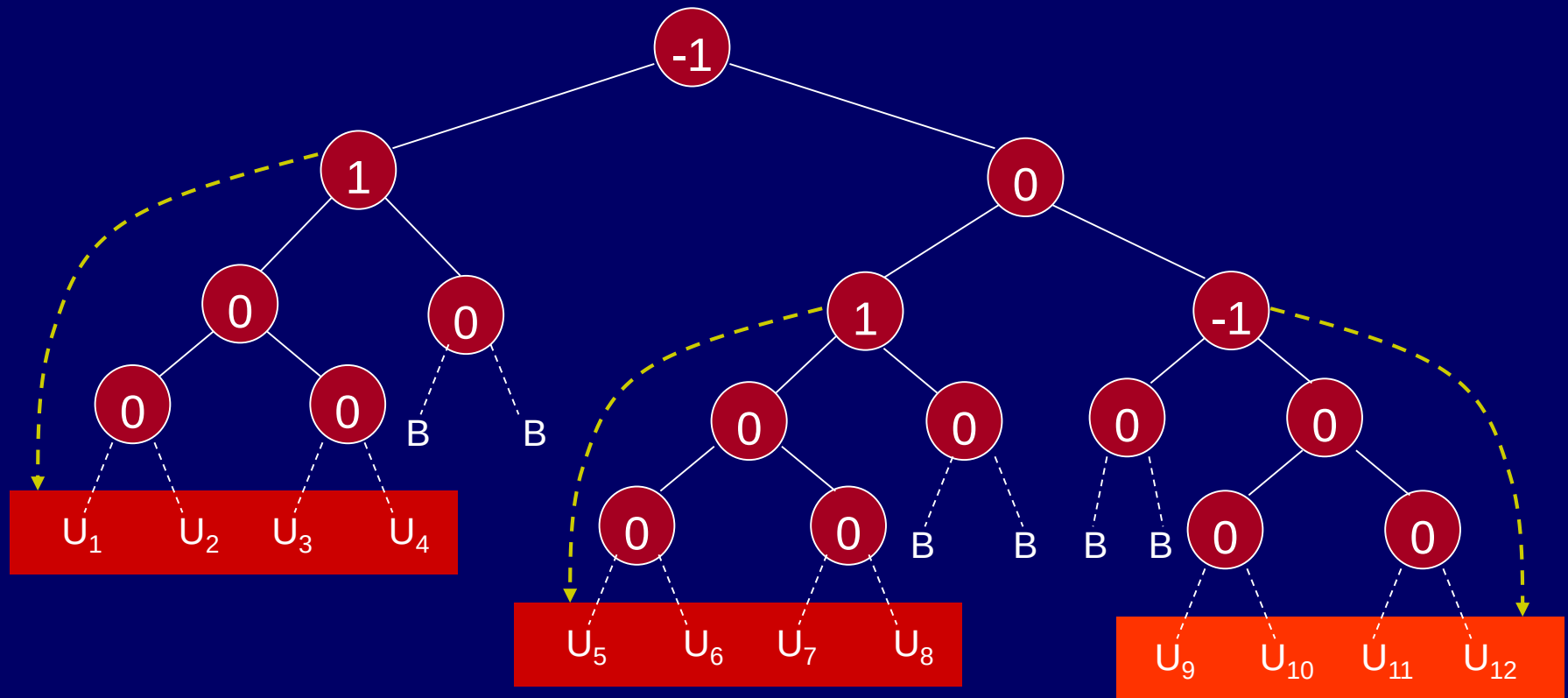
Balanced Binary Tree



- Tree becomes unbalanced only if the newly inserted node
 - is a left descendant of a node that previously had a balance of 1 (U_1 to U_8),
 - or is a descendant of a node that previously had a balance of -1 (U_9 to U_{12})

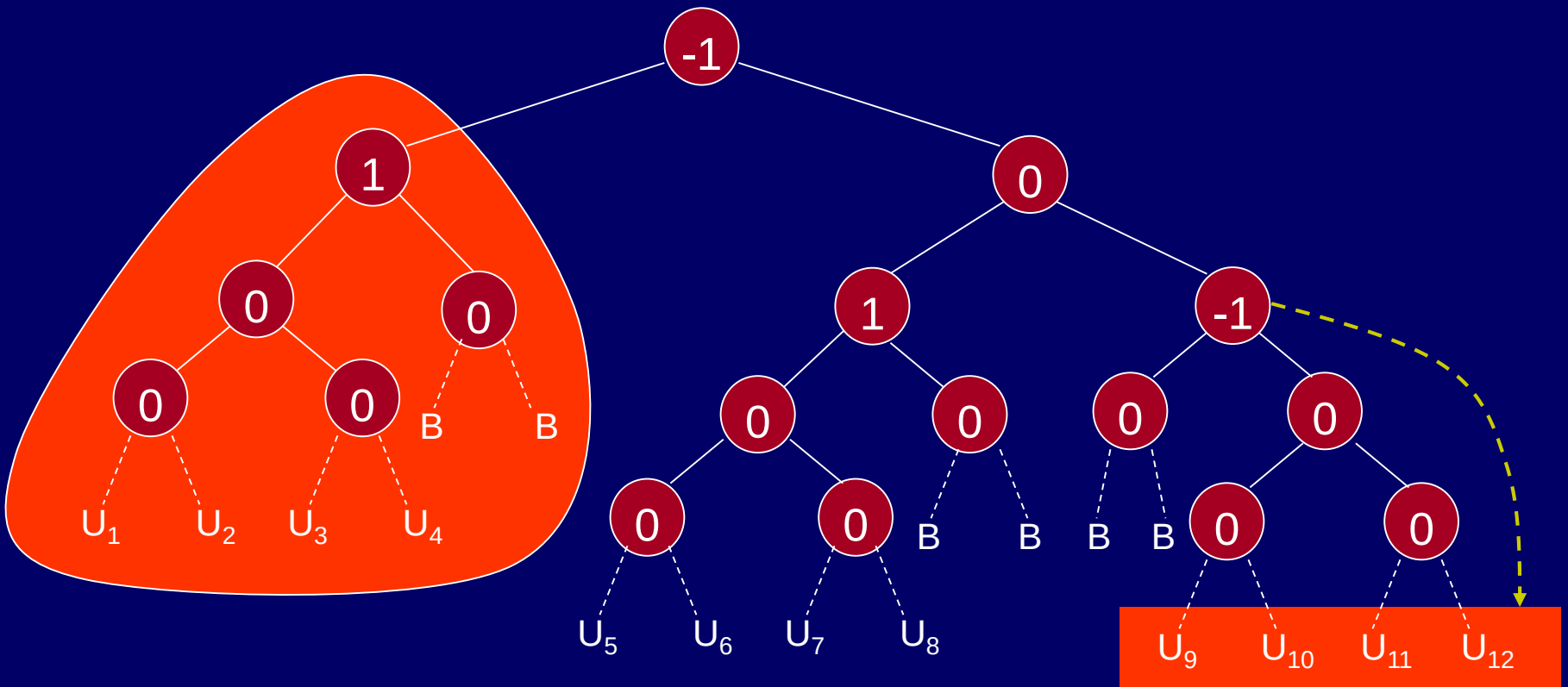
Balanced Binary Tree

Insertions and effect on balance

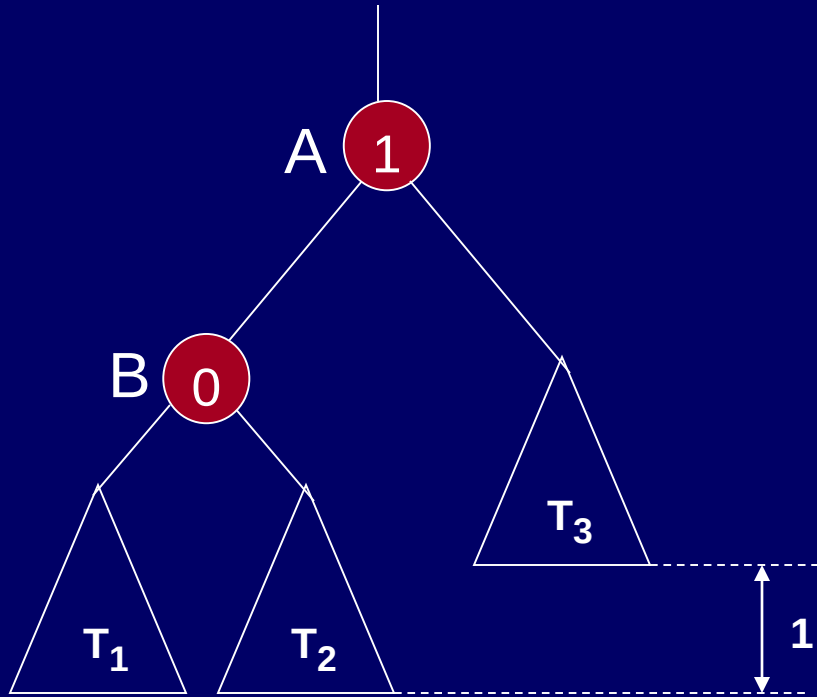


Balanced Binary Tree

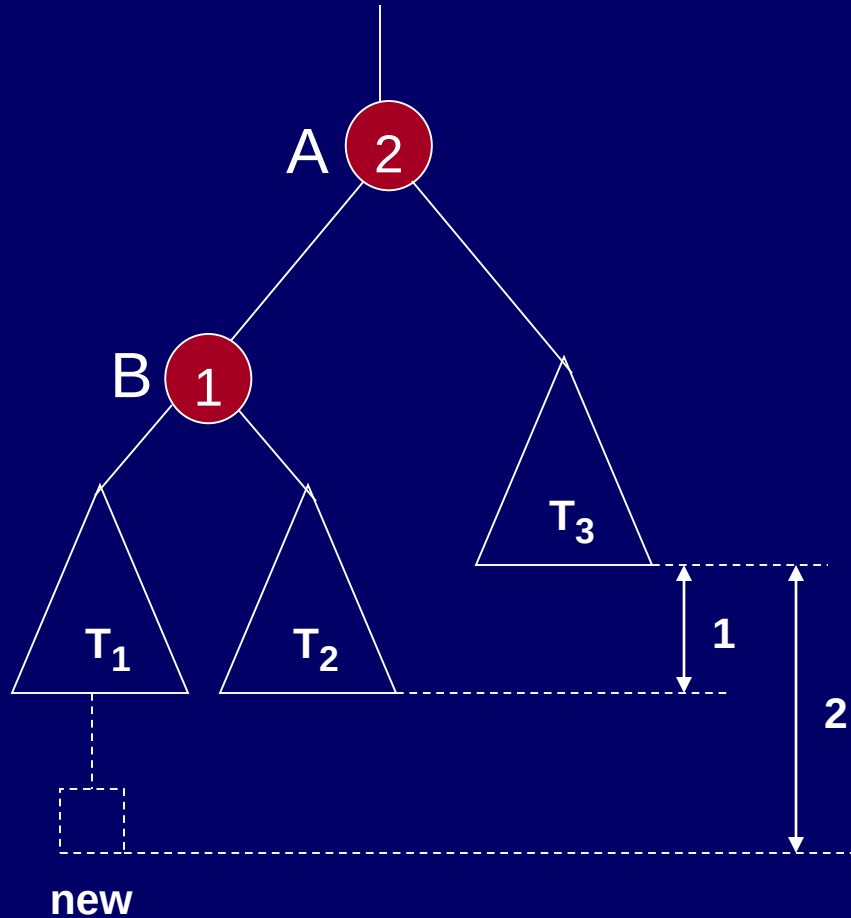
Consider the case of node that was previously 1



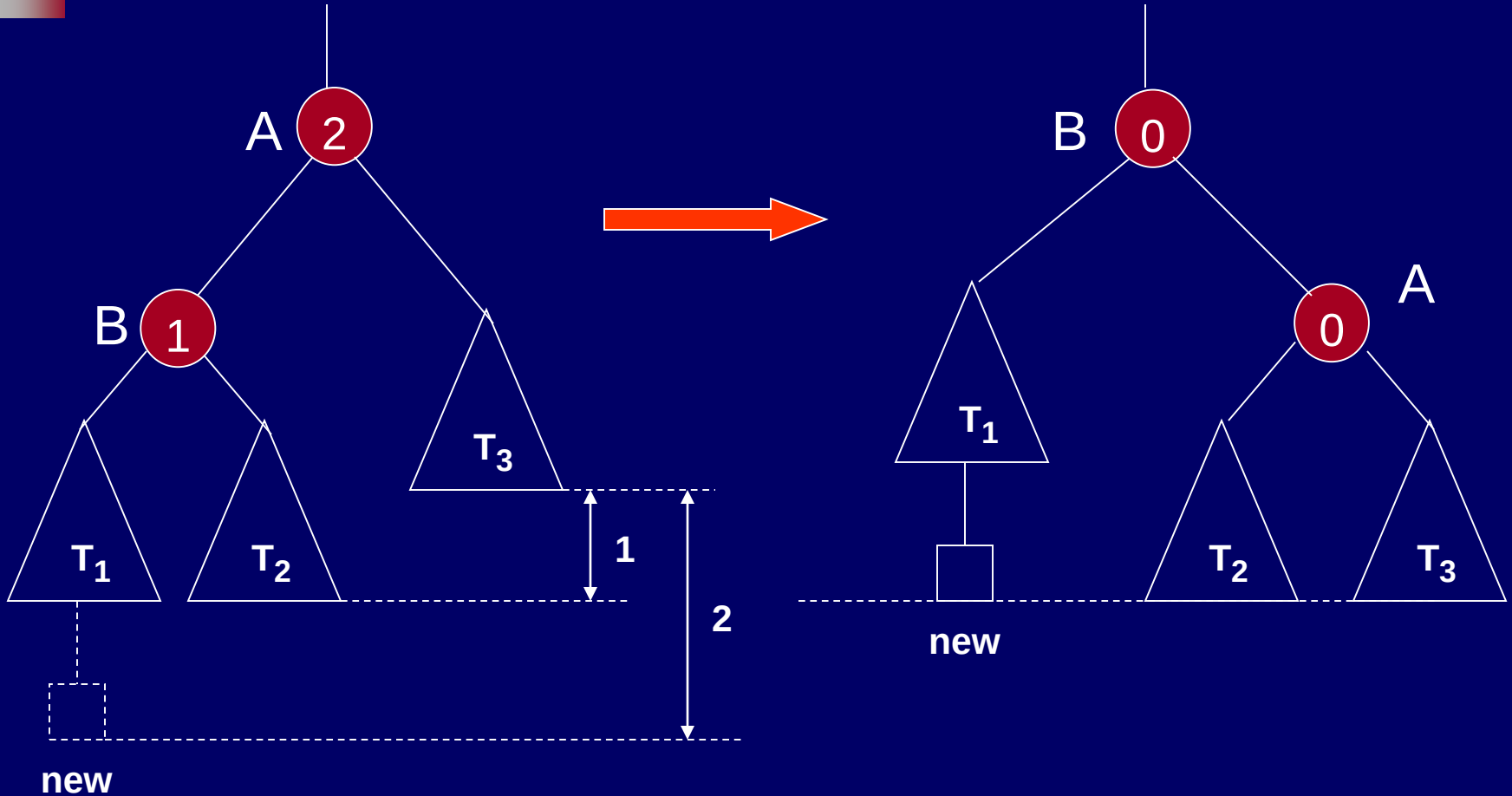
Inserting New Node in AVL Tree



Inserting New Node in AVL Tree



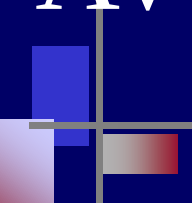
Inserting New Node in AVL Tree



Inorder: T₁ B
T₂ A T₃

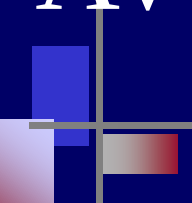
Inorder: T₁ B
T₂ A T₃

AVL Tree Building Example



- Let us work through an example that inserts numbers in a balanced search tree.
- We will check the balance after each insert and rebalance if necessary using rotations.

AVL Tree Building Example



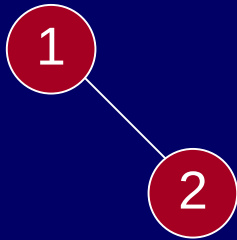
Insert(1)

1

AVL Tree Building Example

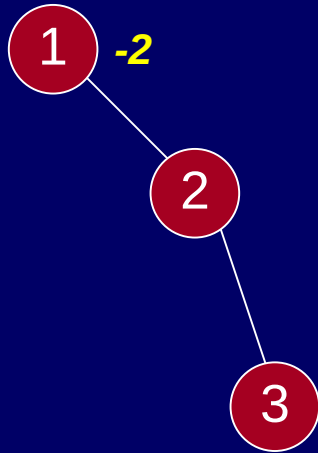


Insert(2)



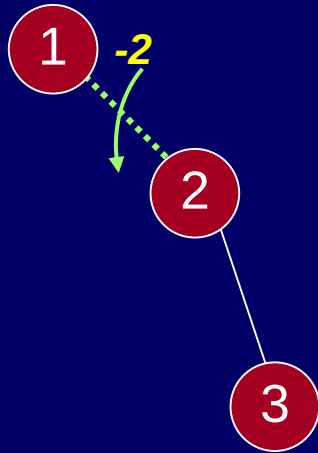
AVL Tree Building Example

Insert(3) single left rotation



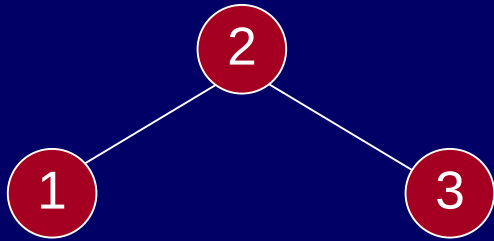
AVL Tree Building Example

Insert(3) single left rotation



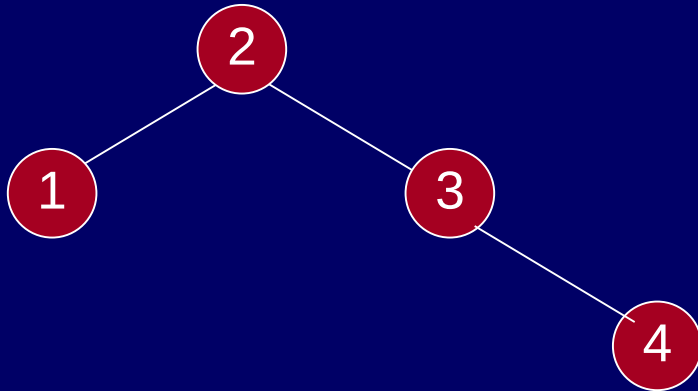
AVL Tree Building Example

Insert(3)



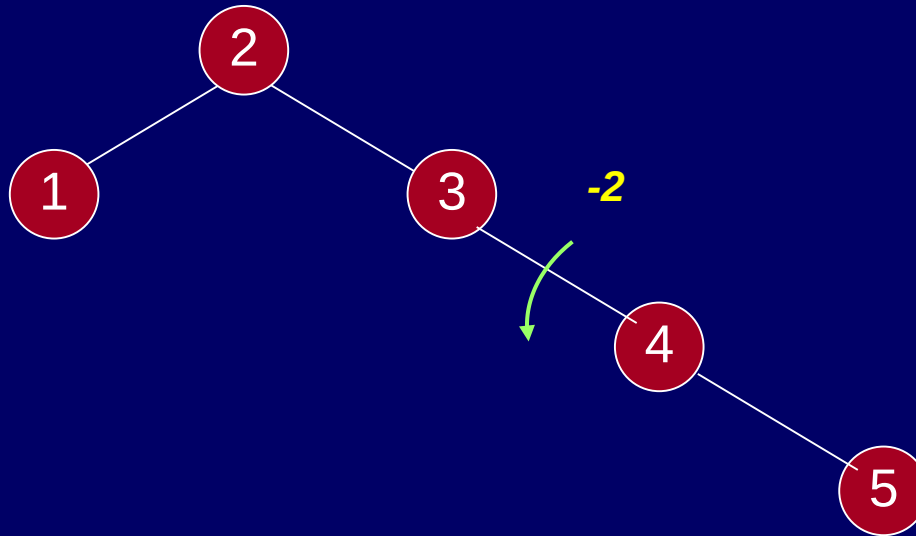
AVL Tree Building Example

Insert(4)



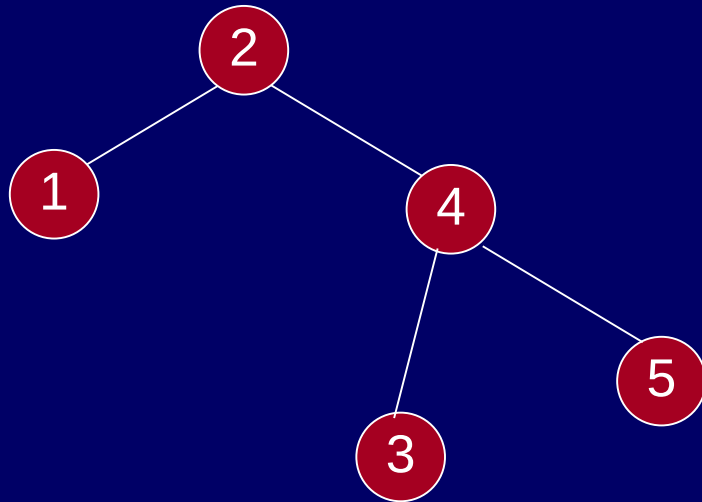
AVL Tree Building Example

Insert(5)



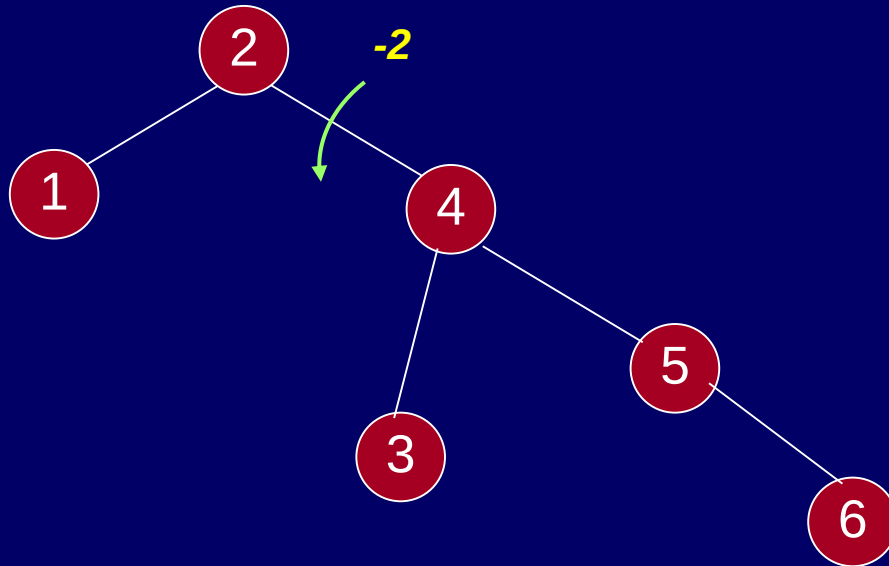
AVL Tree Building Example

Insert(5)



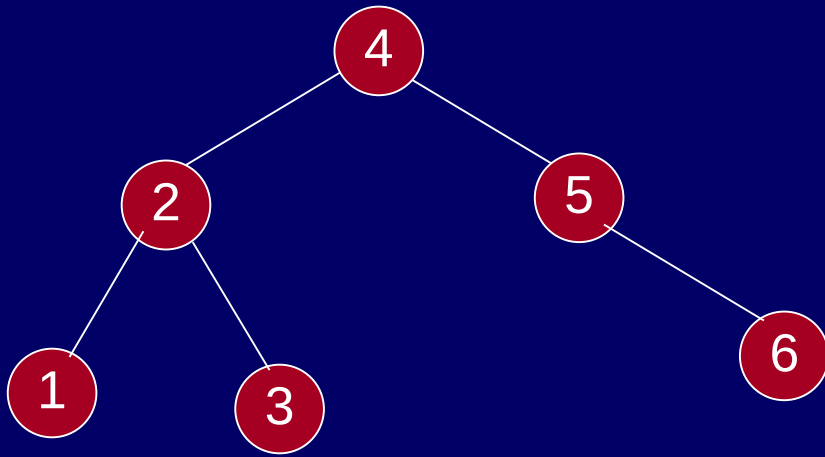
AVL Tree Building Example

Insert(6)



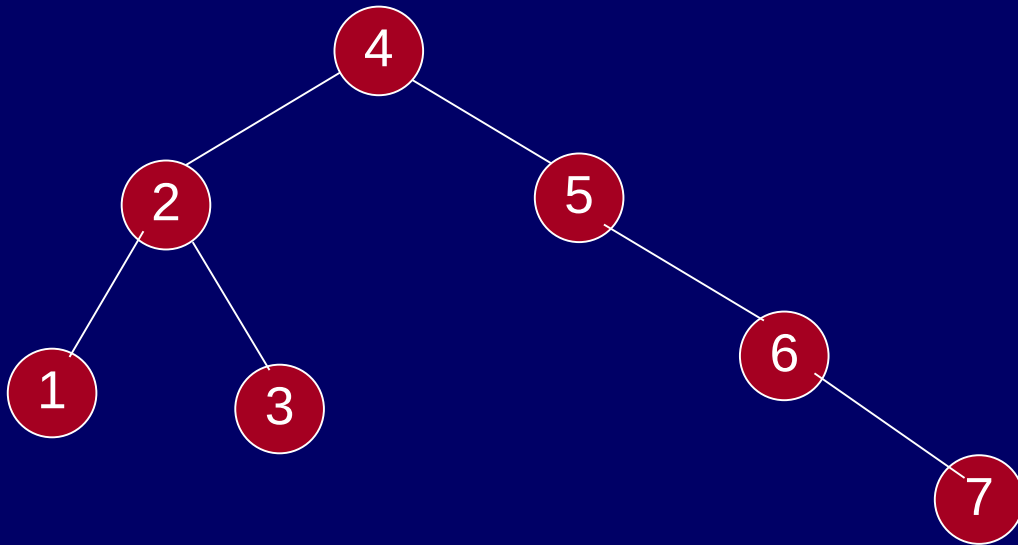
AVL Tree Building Example

Insert(6)



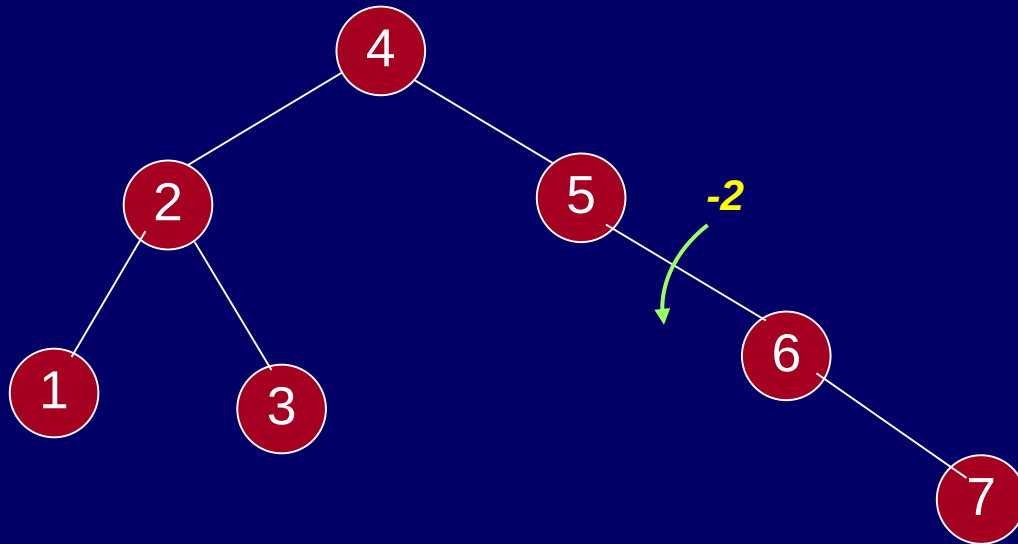
AVL Tree Building Example

Insert(7)



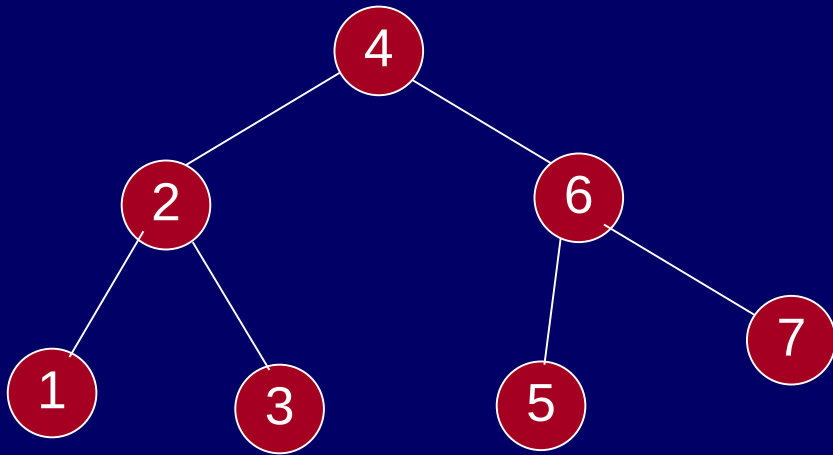
AVL Tree Building Example

Insert(7)



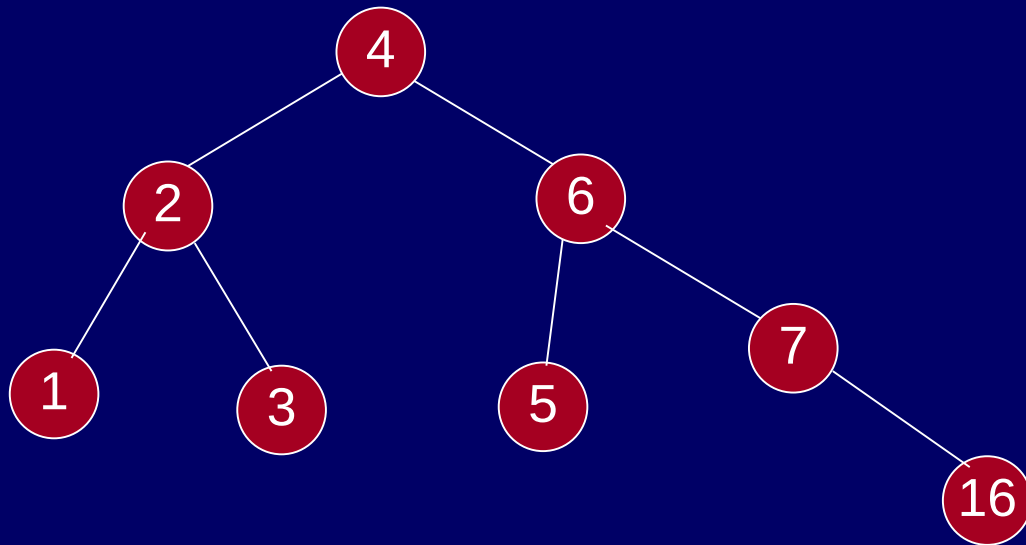
AVL Tree Building Example

Insert(7)



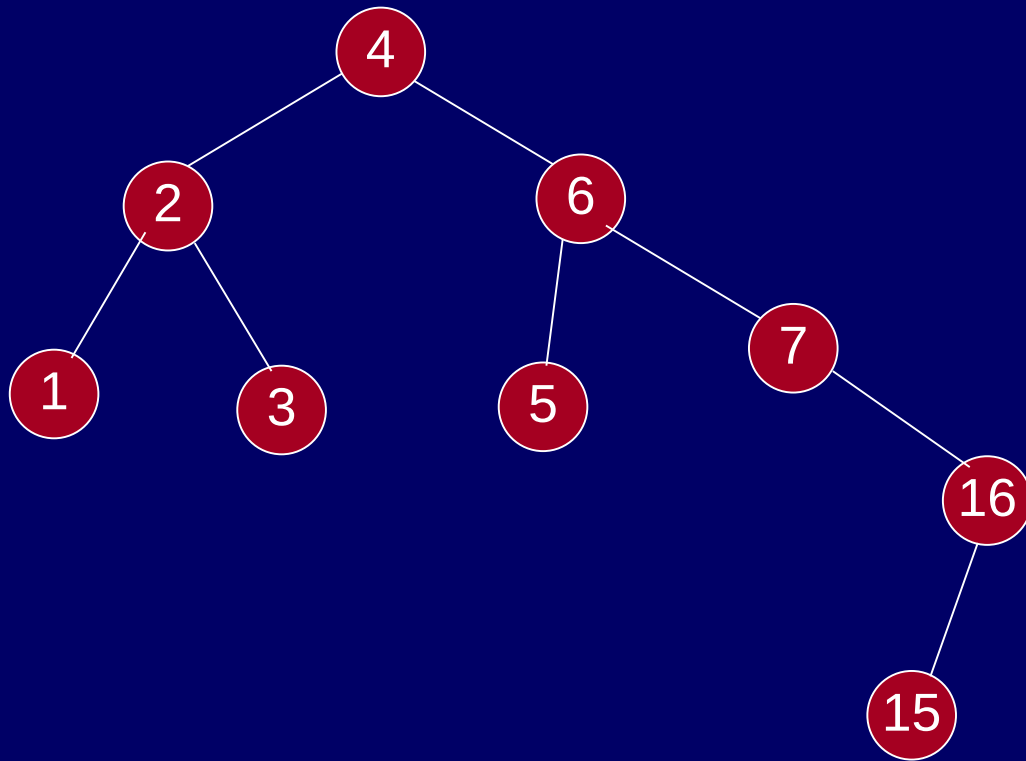
AVL Tree Building Example

Insert(16)



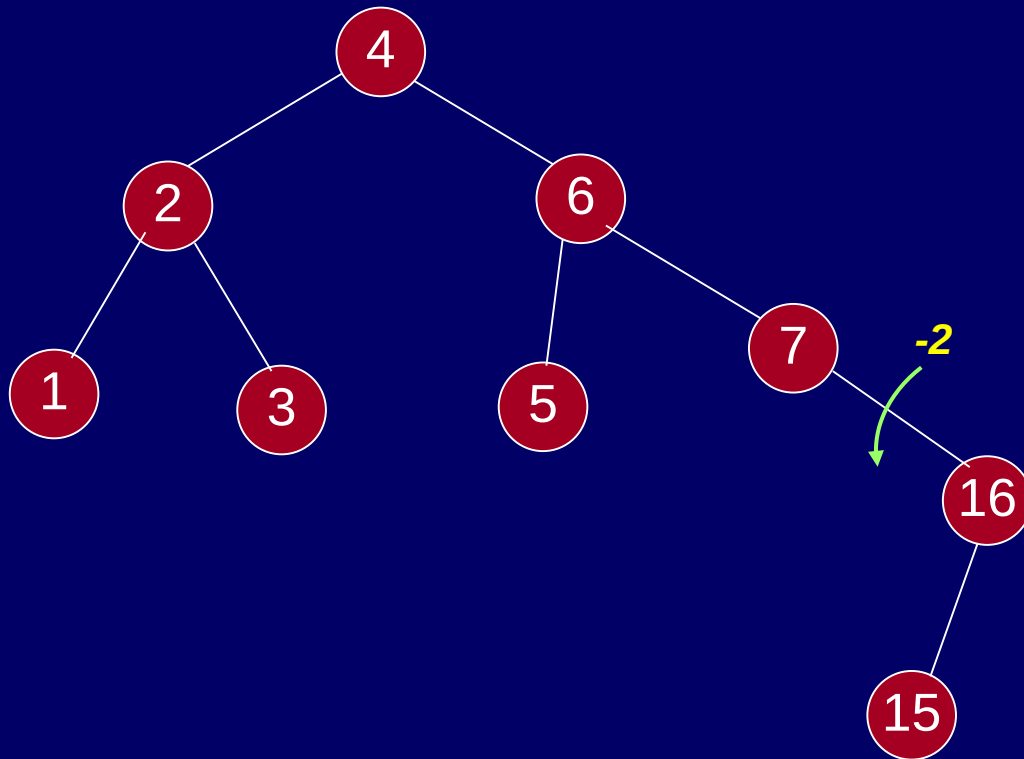
AVL Tree Building Example

Insert(15)



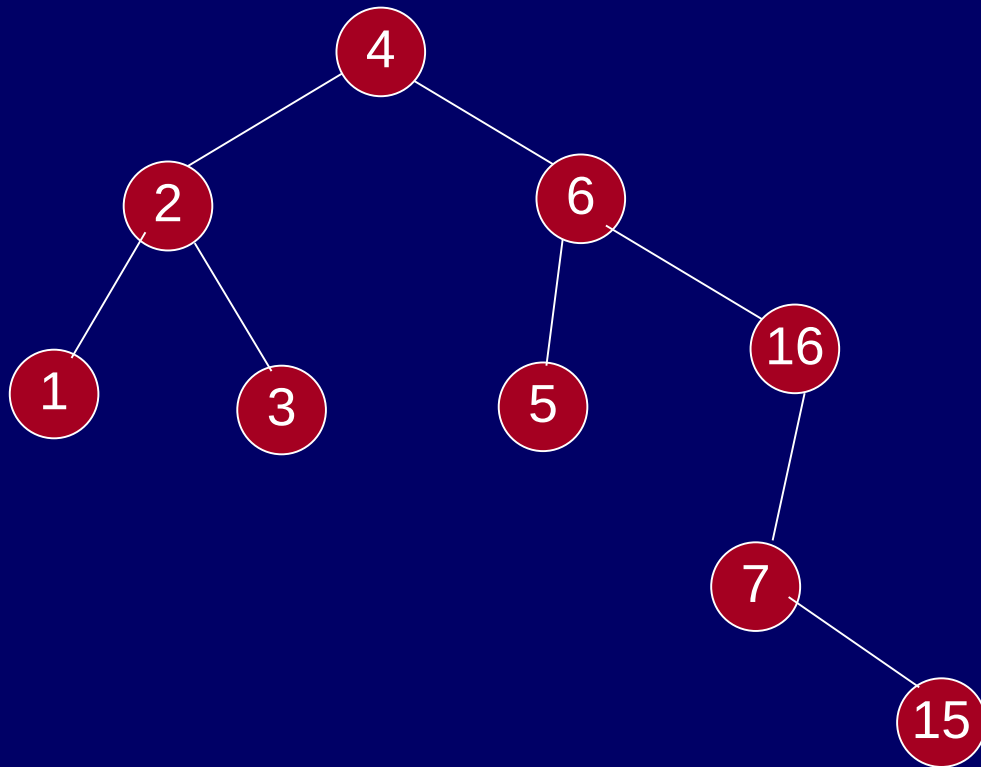
AVL Tree Building Example

Insert(15)



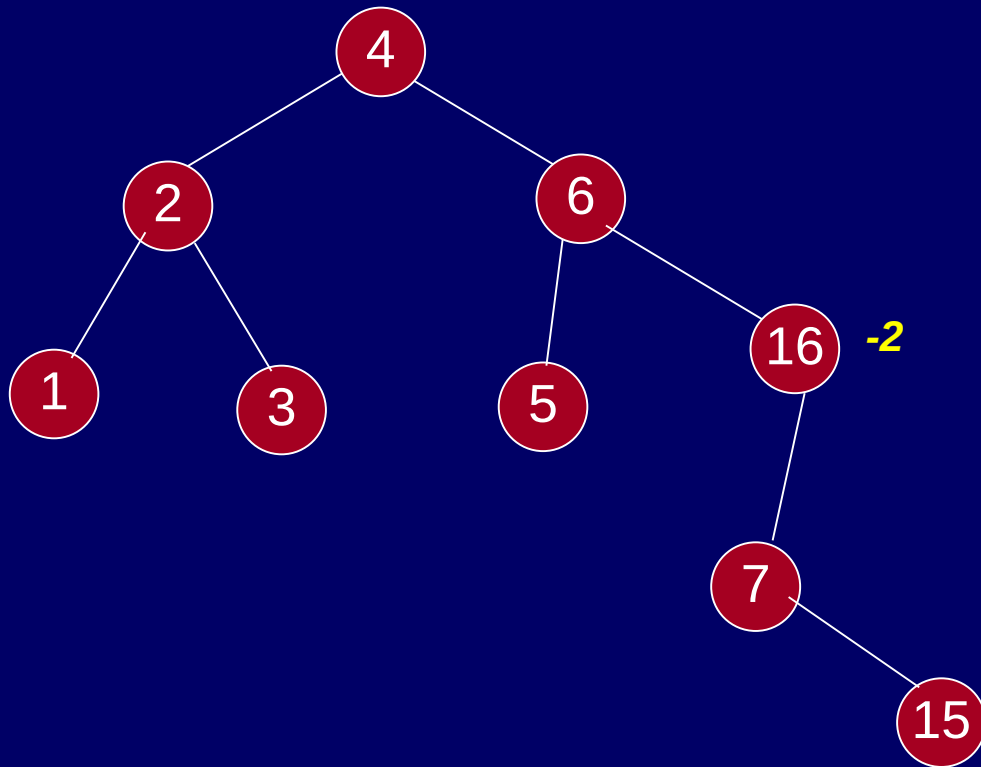
AVL Tree Building Example

Insert(15)

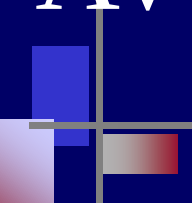


AVL Tree Building Example

Insert(15)

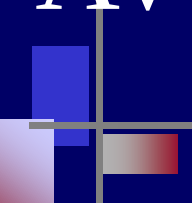


AVL Tree Building Example



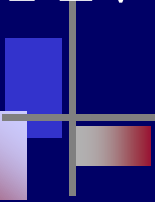
- Single rotation does not seem to restore the balance
- The problem is the node 15 is an inner subtree that is too deep.
- Let us revisit the rotation

AVL Tree Building Example



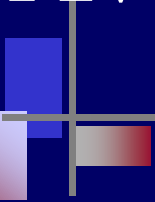
- Let us call the node that must be rebalanced α
- Since any node has at most two children, and height imbalance requires that α 's node two subtrees differ by two (or -2), the violation will occur in **four cases**

AVL Tree Building Example



- i. An insertion into left subtree of the left child of α
- ii. An insertion into right subtree of the left child of α
- iii. An insertion into left subtree of the right child of α
- iv. An insertion into right subtree of the right child of α

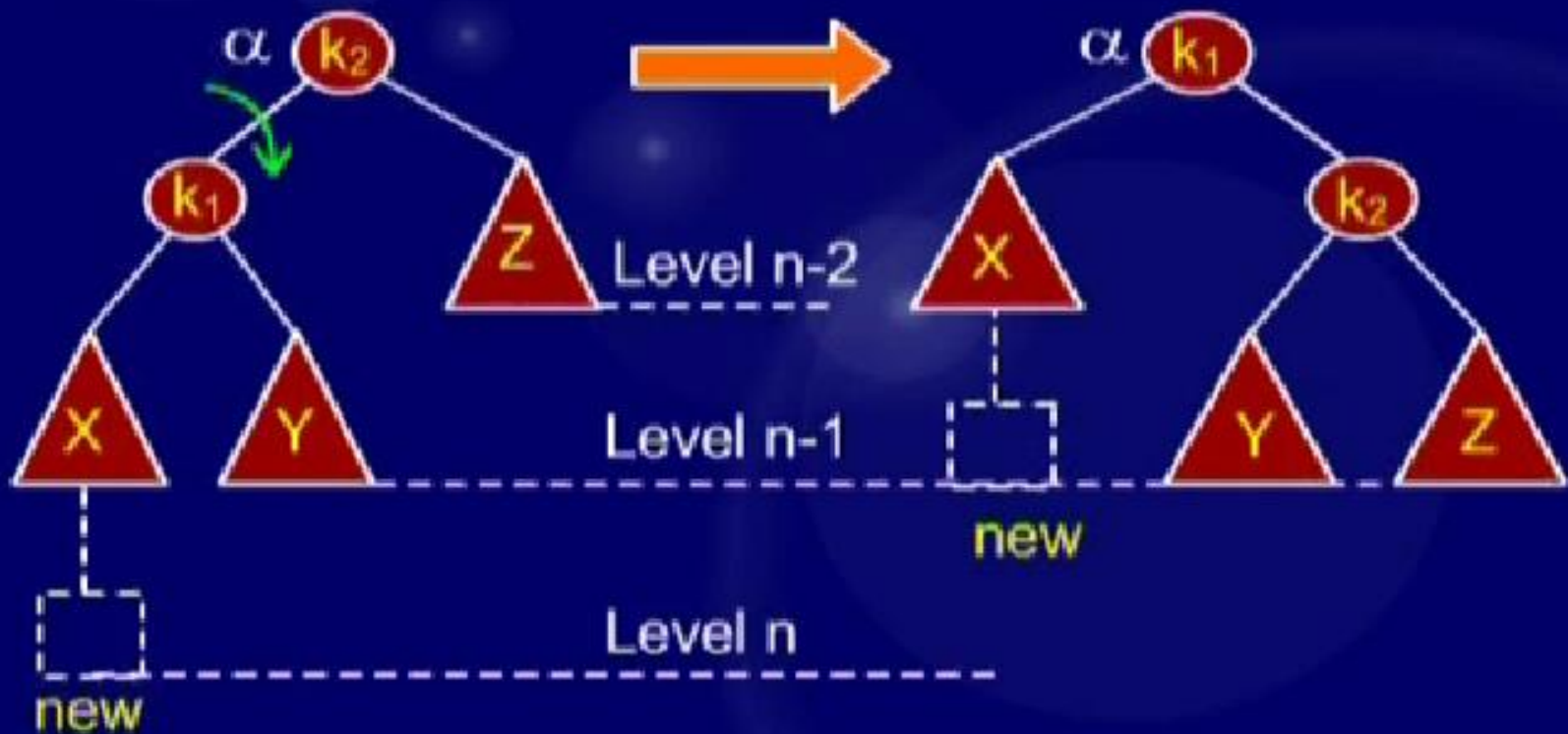
AVL Tree Building Example



- The insertion occurs on the “outside” (i.e., left-left or right-right) in cases 1 and 4.
- Single rotation can fix the balance in cases 1 and 4
- Insertion occurs on the “inside” in cases 2 and 3 which single rotation cannot fix.

AVL Tree Building Example

Single right rotation to fix case 1.



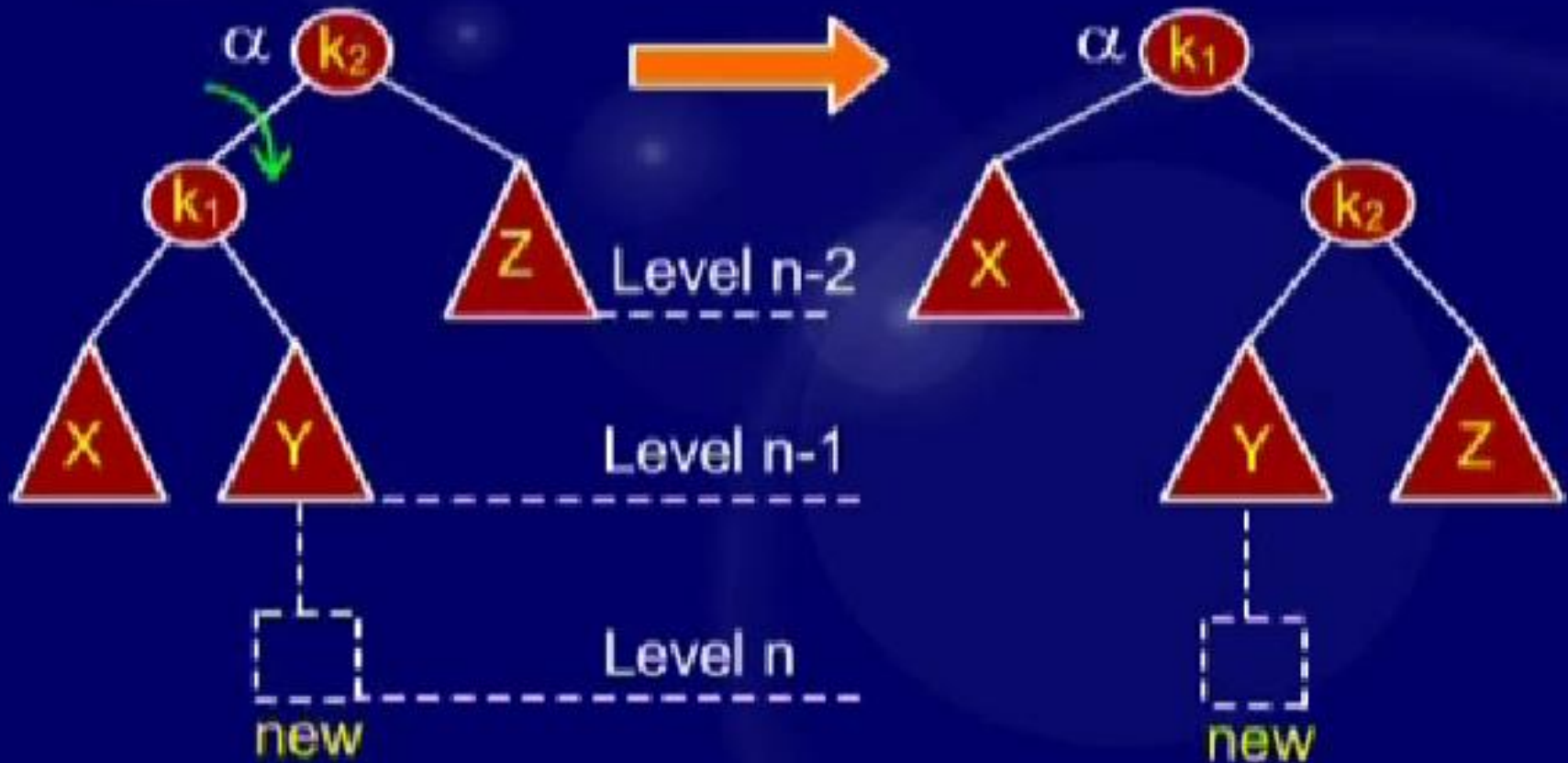
AVL Tree Building Example

Single left rotation to fix case 4.

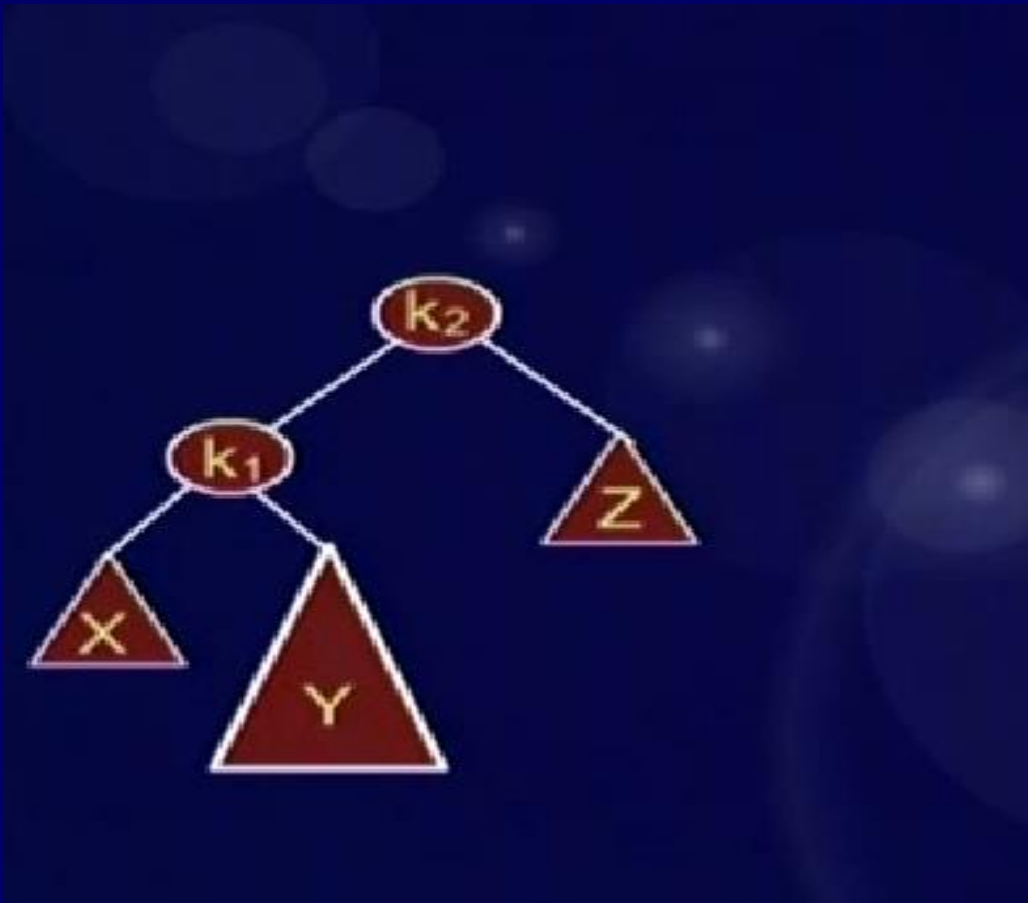


AVL Tree Building Example

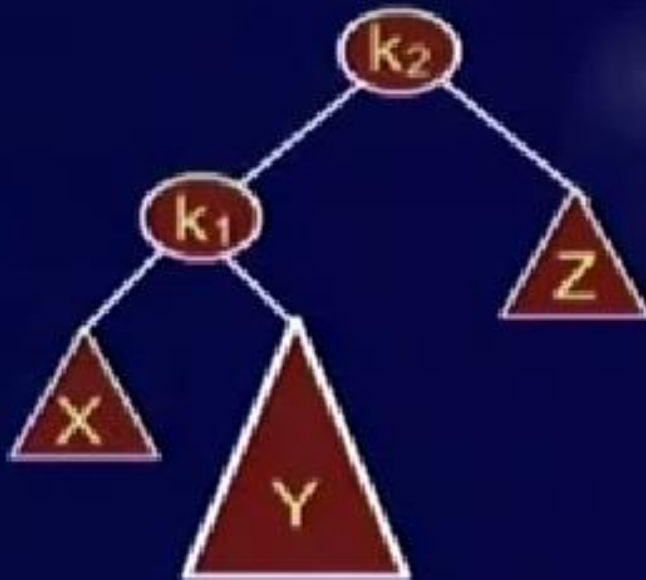
Single right rotation *fails* to fix case 2.



AVL Tree Building Example

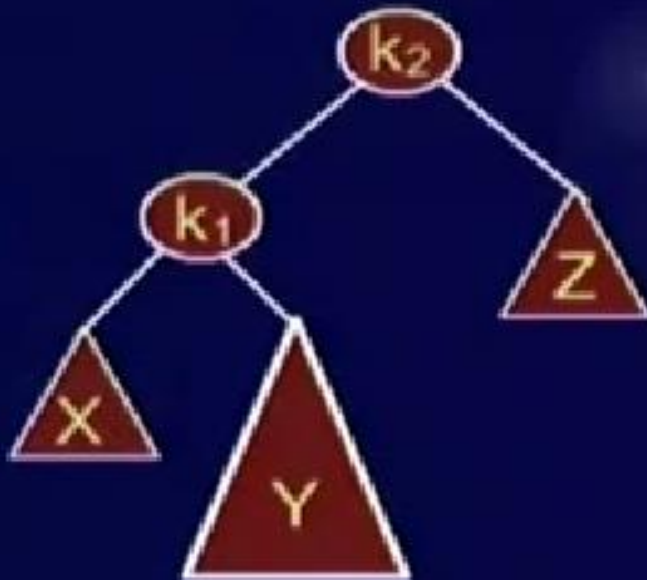


AVL Tree Building Example



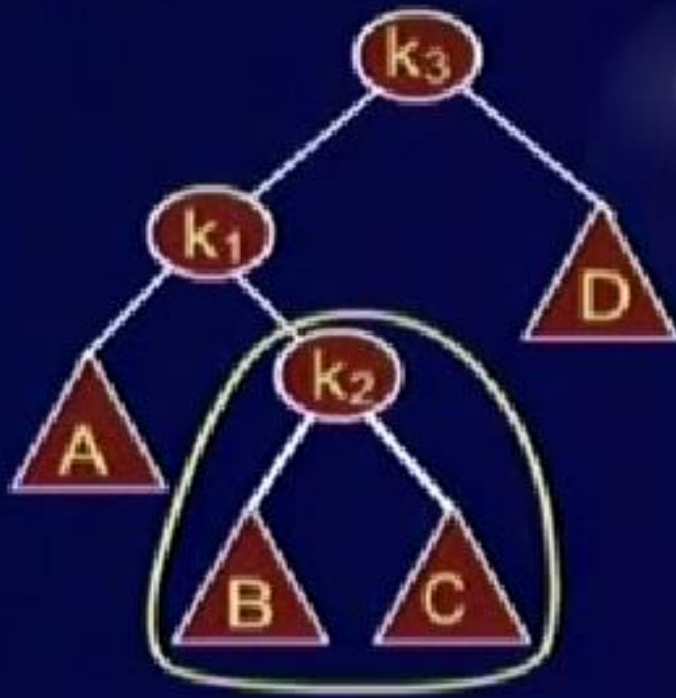
- Y is non-empty because the new node was inserted in Y.
- Thus Y has a root and two subtrees.
- View the entire tree with four subtrees connected with 3 nodes.

AVL Tree Building Example



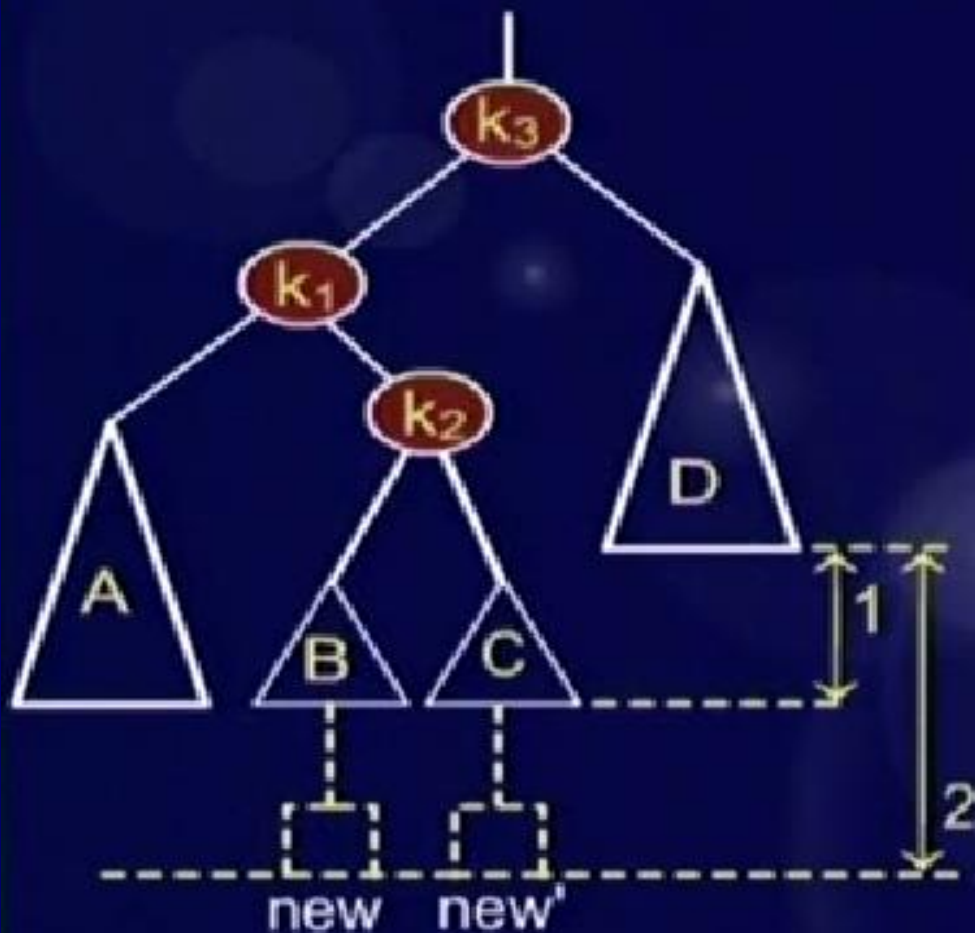
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AVL Tree Building Example



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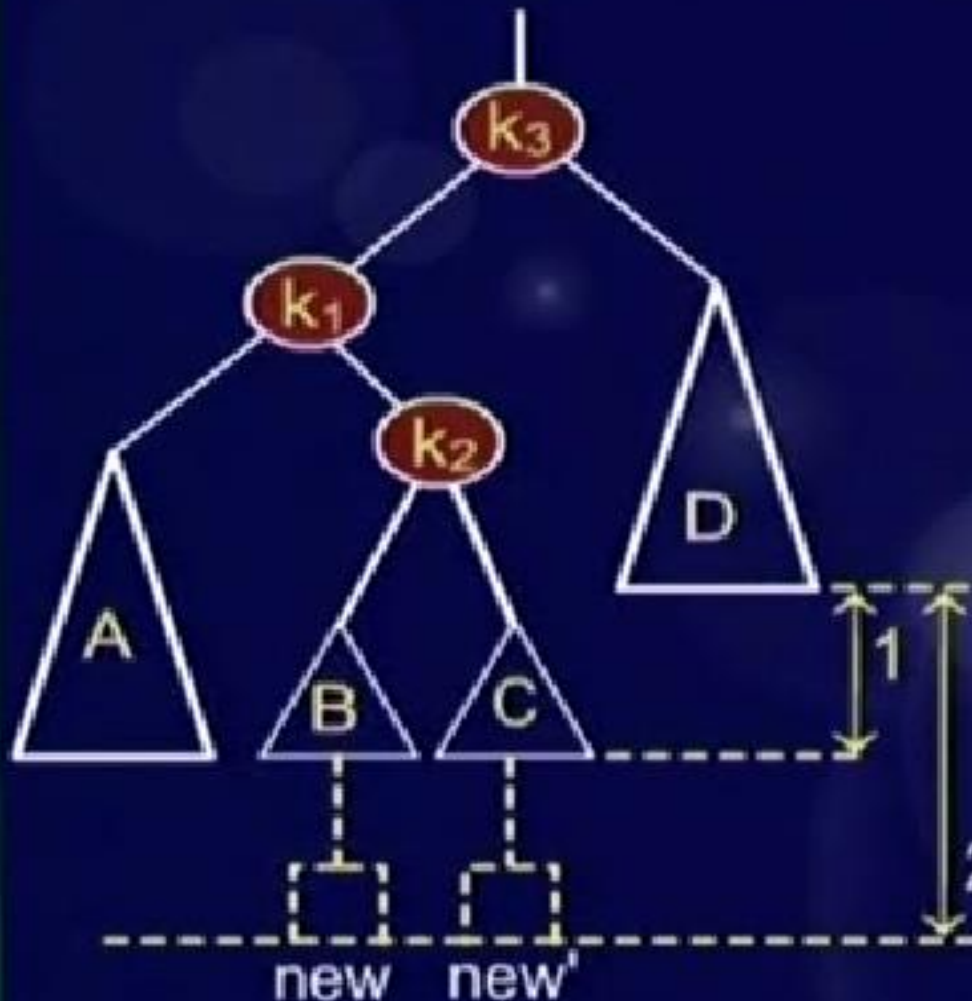
AVL Tree Building Example



- Exactly one of tree B or C is two levels deeper than D; we are not sure which one.

New node inserted at either of the two spots

AVL Tree Building Example

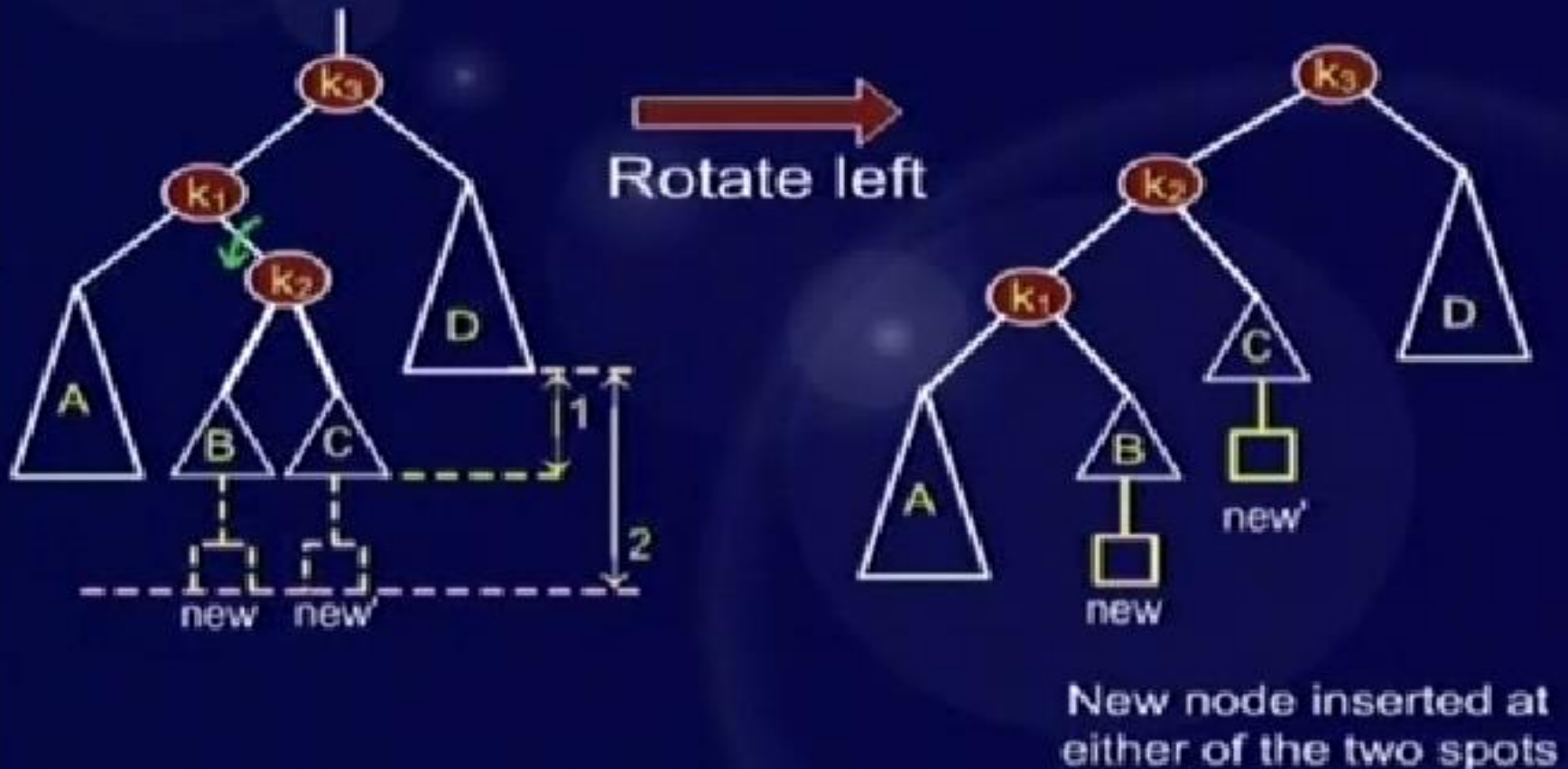


New node inserted at either of the two spots

- A rotation between k_3 and k_1 (k_3 was k_2 then) was shown not to work.
- The only alternative is to place k_2 as the new root.
- This forces k_1 to be k_2 's left child and k_3 to be its right child.

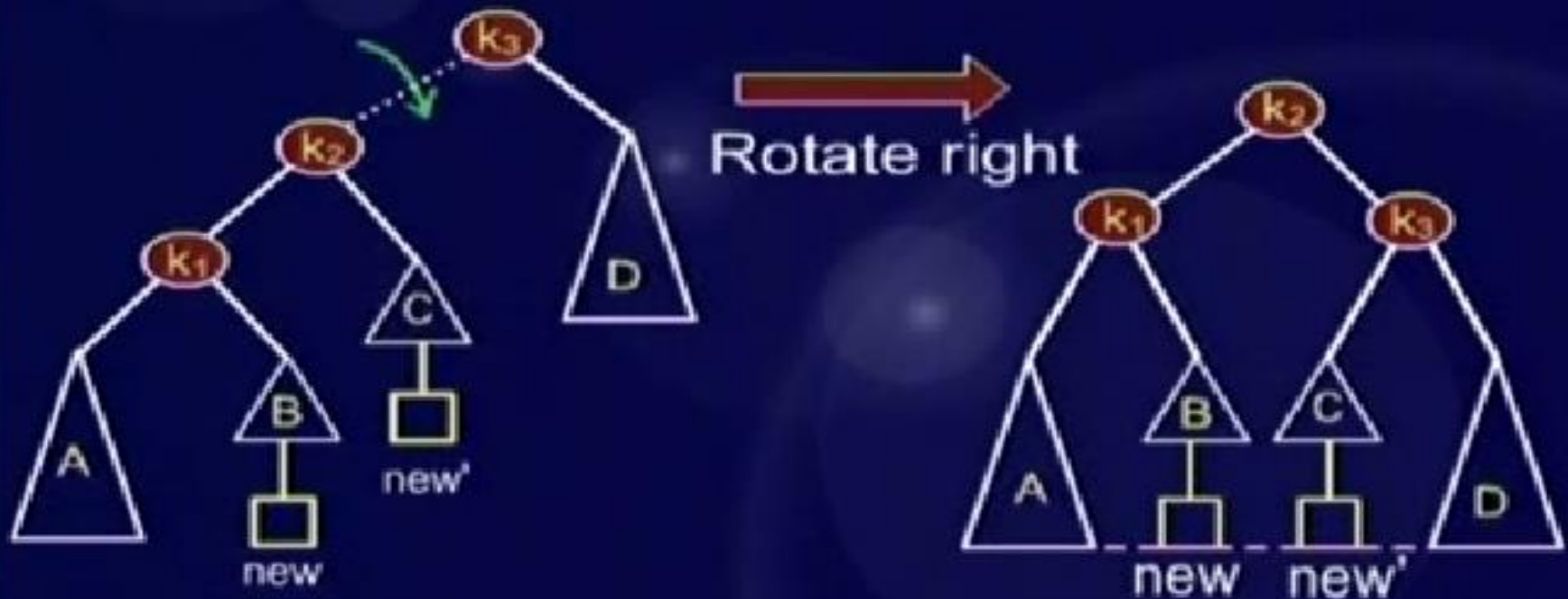
AVL Tree Building Example

- Left-right **double** rotation to fix case 2.



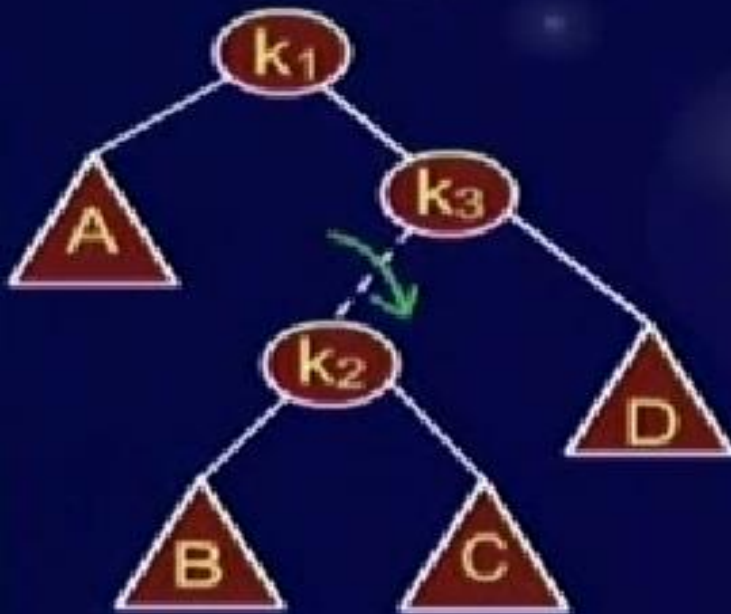
AVL Tree Building Example

- Left-right **double** rotation to fix case 2.



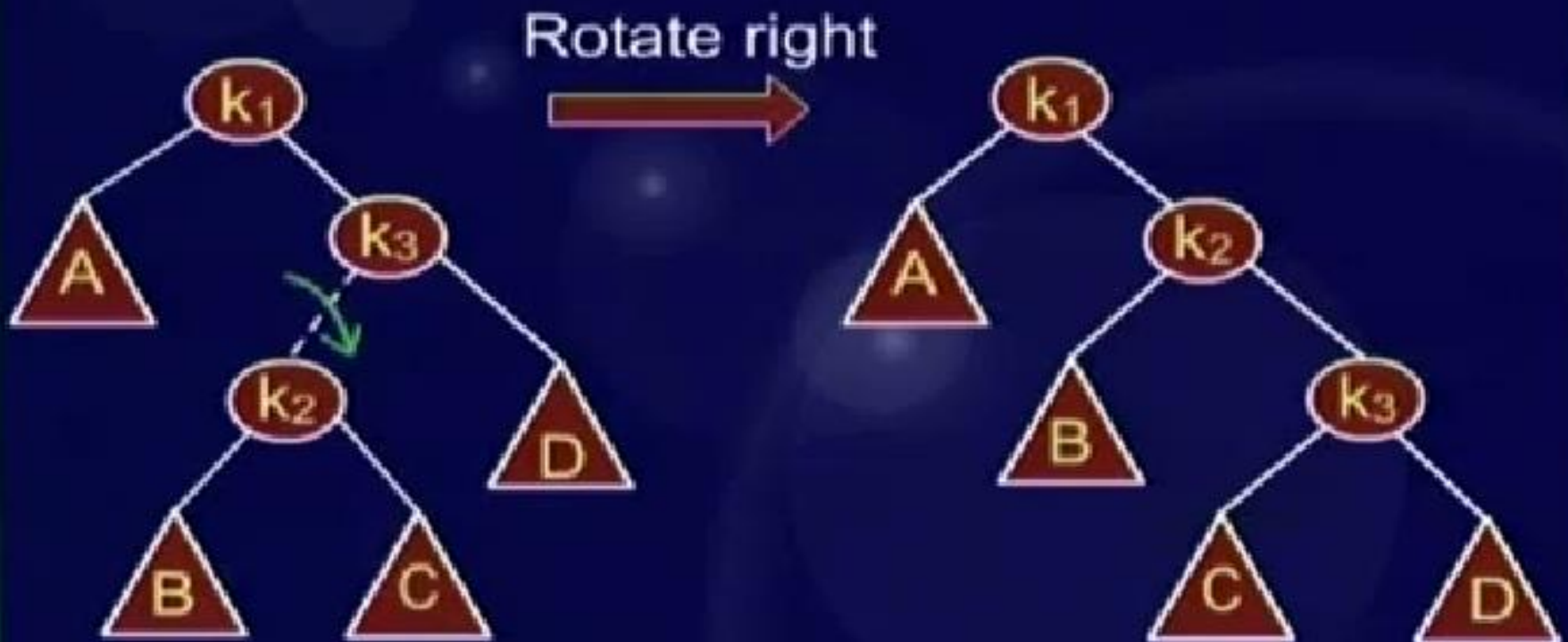
AVL Tree Building Example

- Right-left **double** rotation to fix case 3.



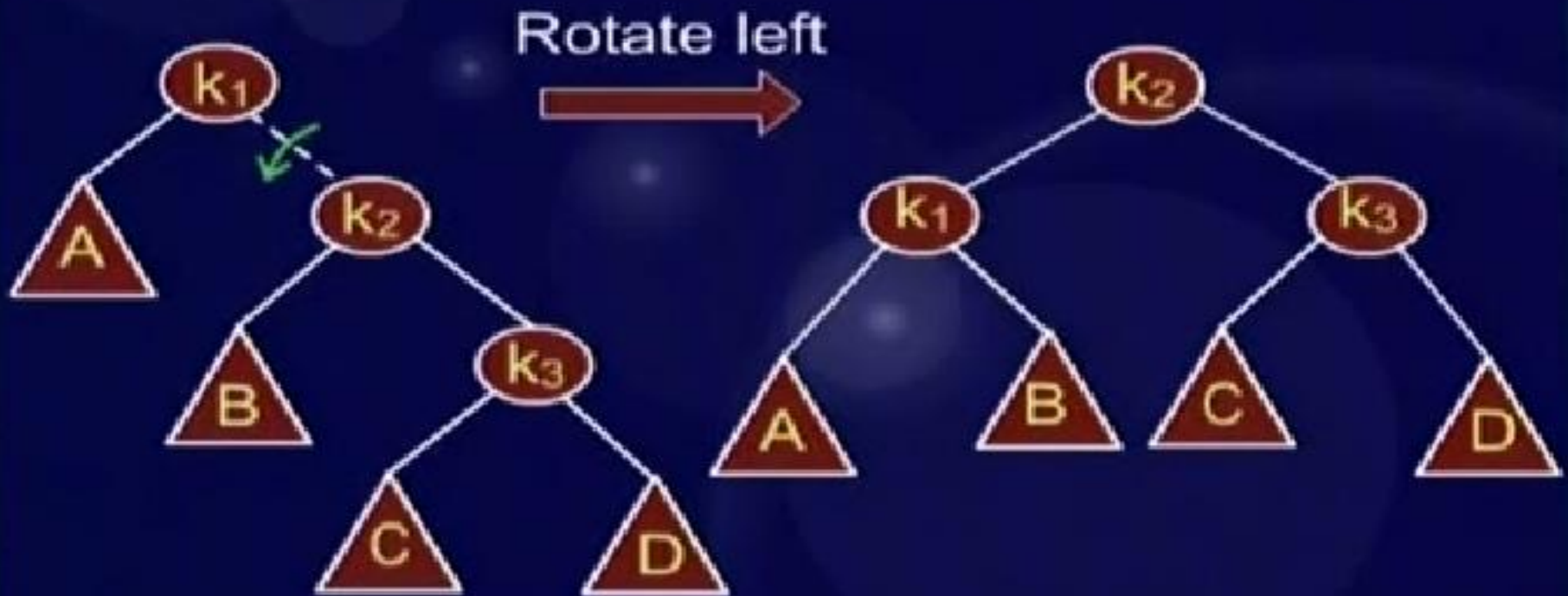
AVL Tree Building Example

- Right-left **double** rotation to fix case 3.



AVL Tree Building Example

- Right-left **double** rotation to fix case 3.



AVL Tree Building Example

Insert(15)



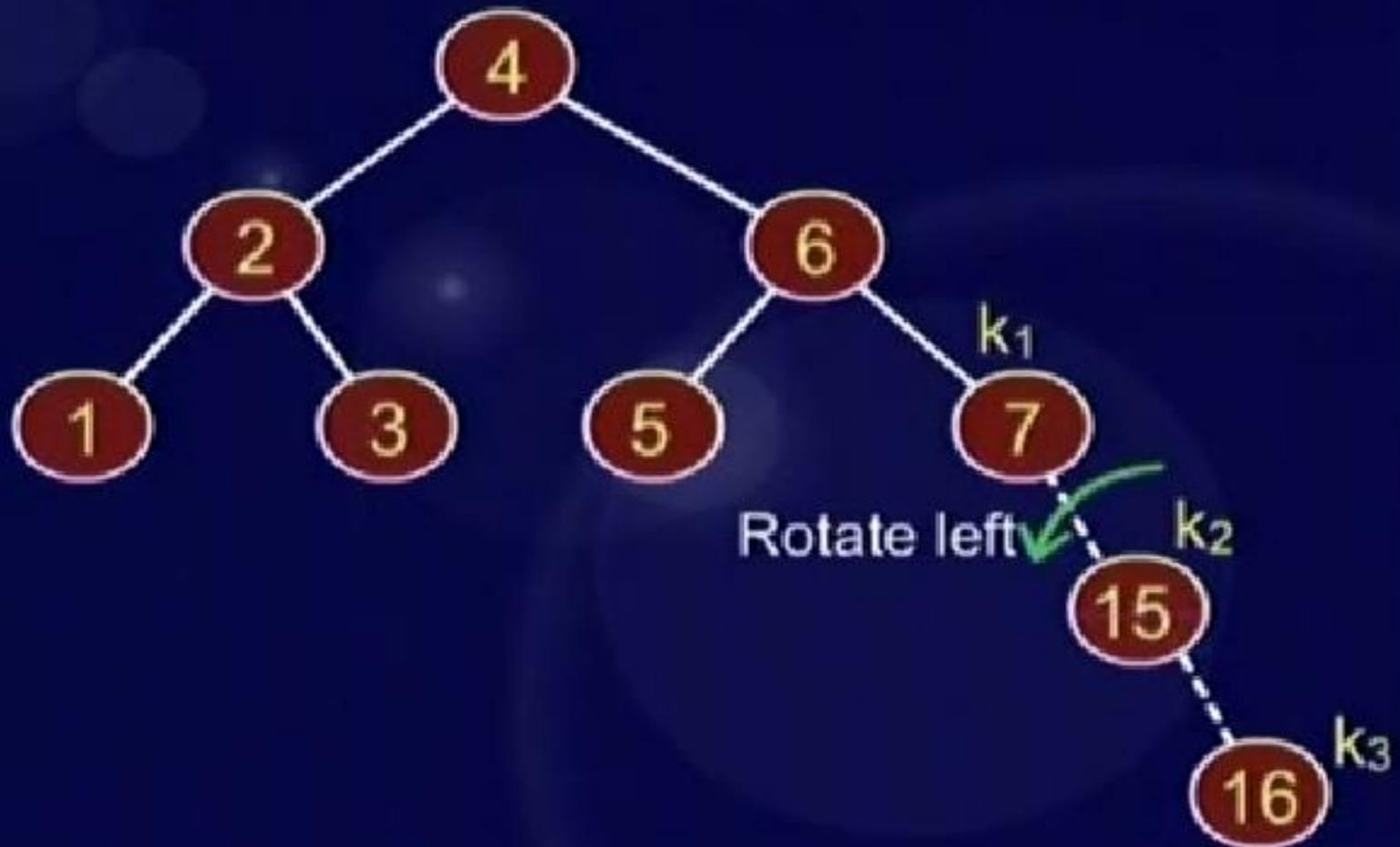
AVL Tree Building Example

Insert(15) right-left double rotation



AVL Tree Building Example

Insert(15) right-left double rotation



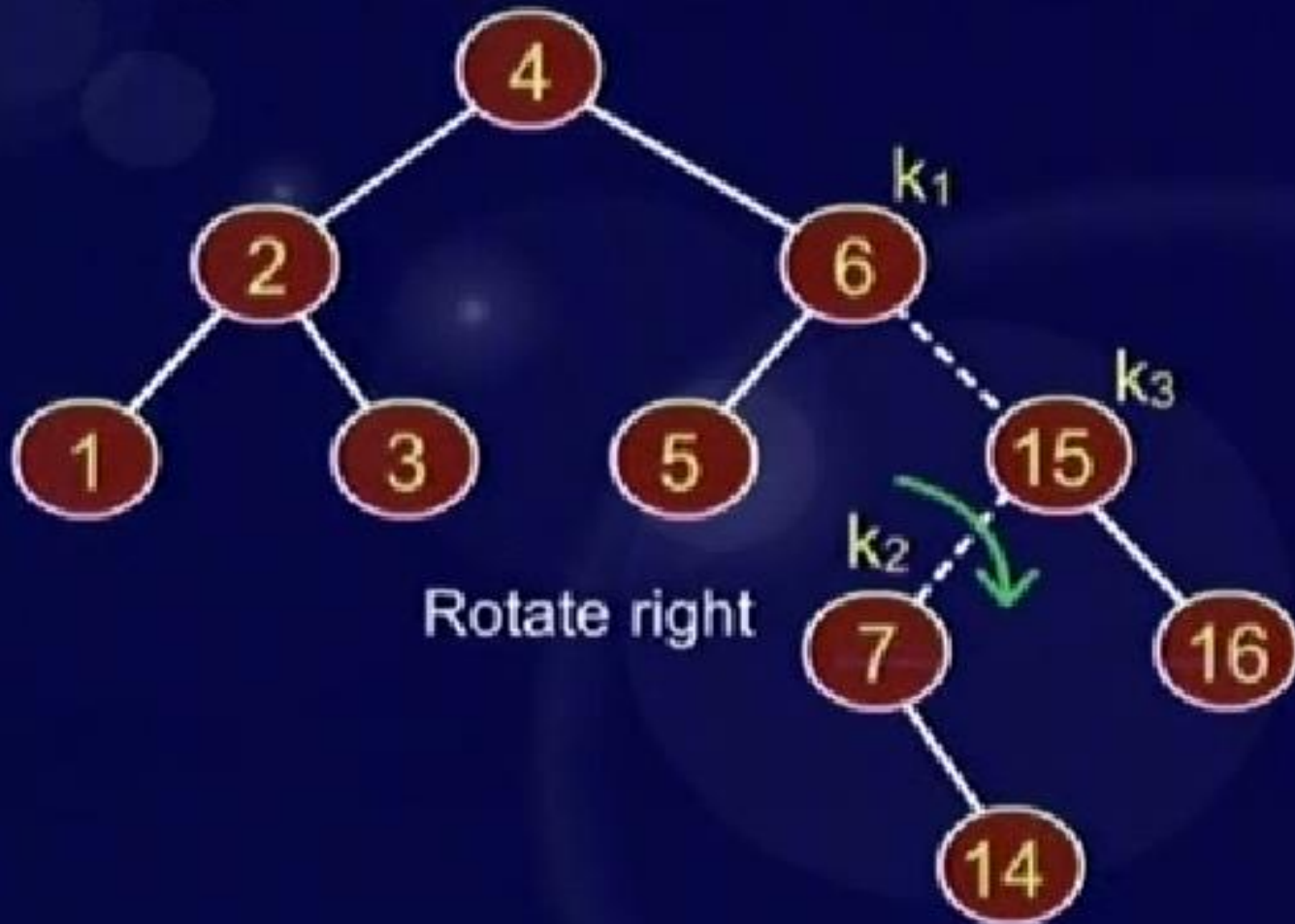
AVL Tree Building Example

Insert(15) right-left double rotation



AVL Tree Building Example

Insert(14): right-left double rotation



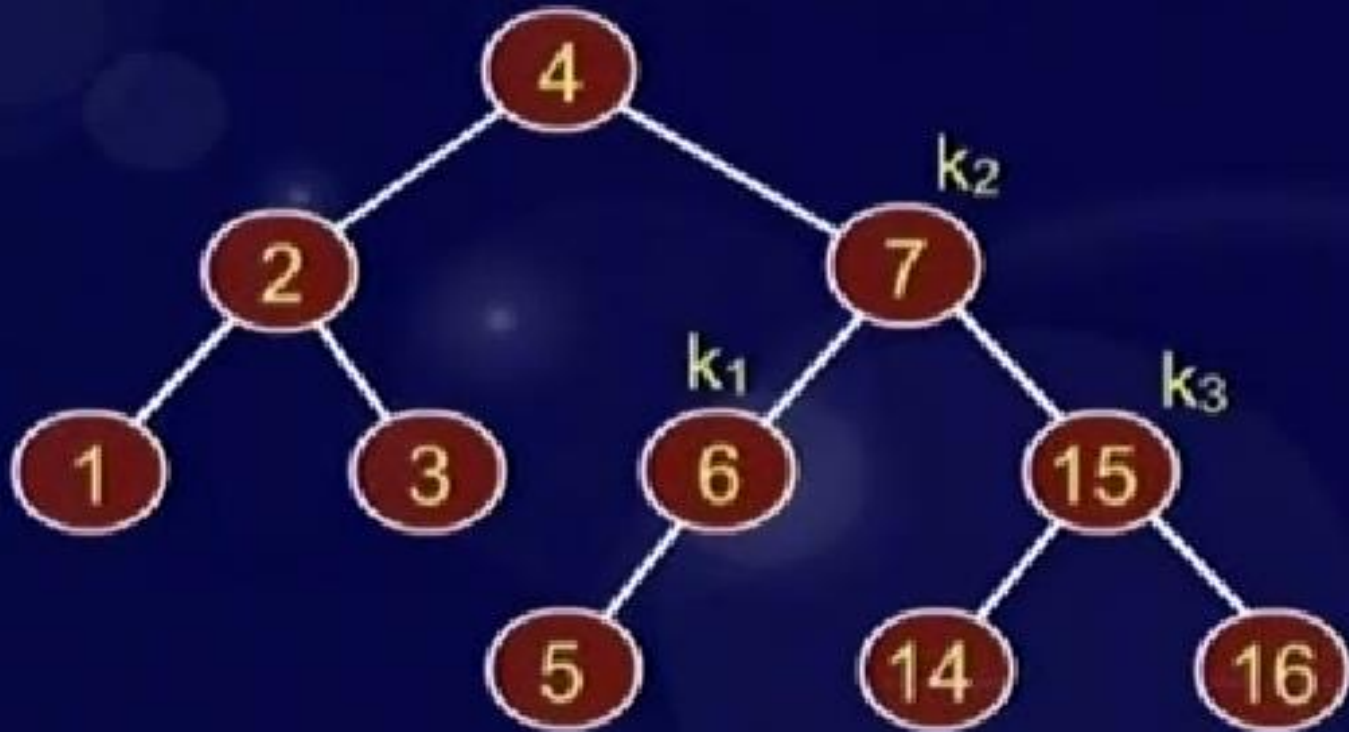
AVL Tree Building Example

Insert(14): right-left double rotation



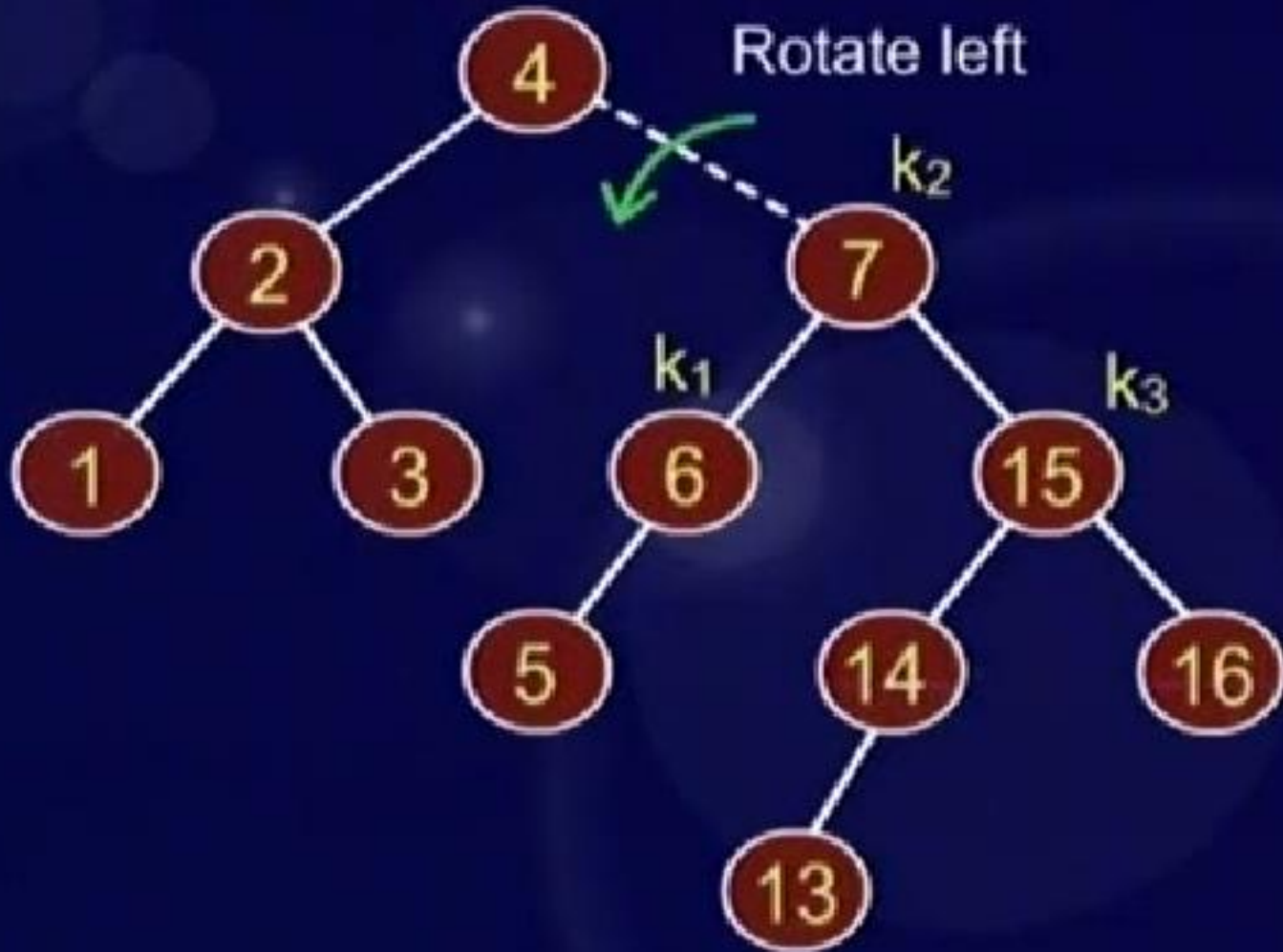
AVL Tree Building Example

Insert(14): right-left double rotation



AVL Tree Building Example

Insert(13): single rotation



AVL Tree Building Example

Insert(13): single rotation

